



Verified Carbon Standard

KA'A PYAHU

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Prepared by	BIOFIX CONSULTORIA LTDA.

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1 PROJECT DETAILS

1.1 Summary Description of the Project

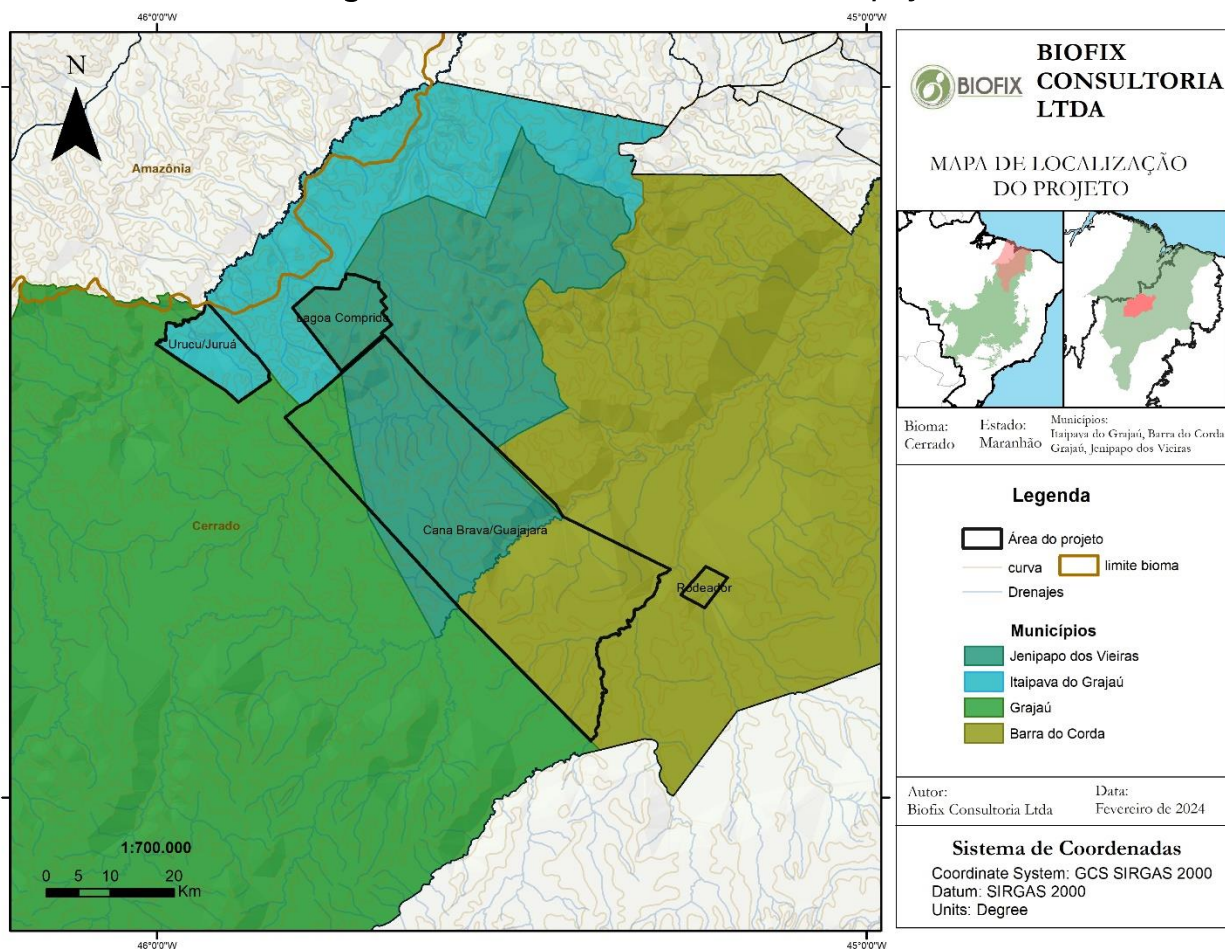
The KA'A PYAHU REDD+ project is located in the state of Maranhão, in the northeast region of Brazil, under the jurisdiction of the Legal Amazon. The proponents of the project are the Guajajara people, inhabitants and holders of three indigenous lands (TI) and one indigenous domain area (DI), namely: Cana Brava Indigenous Land, Urucu-Juruá Indigenous Land, Rodeador Indigenous Dominion Area and Lagoa Comprida Indigenous Land. The Indigenous Land is the traditionally used space by the indigenous peoples, of form constant, for its productive activities and for to preserve the environmental resources necessary for their well-being and physical and cultural reproduction. Domain Lands are properties of indigenous communities, acquired according to foreseen in civil legislation. The Canabrava Indigenous Land and the Area Domain Indigenous Rodeador find in Barra do Corda municipality, the Earth Indigenous Lagoa Comprida is in Jenipapo dos Vieira municipality, and the Earth Urucu/Juruá indigenous group covers the municipality of Grajaú, all in Maranhão, northeastern Brazil.

These territories are historically marked by risk factors such as deforestation, illegal logging that threatens the integrity of the territory and the people who live there, and socio-environmental impacts due to the installation of power lines within the territories. These aspects cause harm to the way of life of these people, who need greater attention to health, education, protection and territorial monitoring. The Ka'a Pyahu project aims to reinforce the aspects covered in the REDD+ investment lines to bring benefits to communities. According to what was discussed in the workshops, the project will recover degraded areas, monitor illegal invasions and control fires. In addition, it will support the production of articles made from tucum straw, an abundant species in the region in the green business line, and basic sanitation to alleviate problems such as the lack of drinking water, among others that are extremely important and directly impact people's lives.

In this way, BIOFIX and the Guajajara of the four territories under the governance of the indigenous association Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara aim to promote the improvement and well-being of the people, in accordance with the priorities listed in the workshops by Guajajara leaders. The project, which is in agreement with the leaders and the association, respects Guajajara cultural principles and is premised on good environmental and social management of the territories. In the assembly, lines of action were established that aim, on the one hand, to meet the most urgent needs of the communities and, on the other, to offer tools for preservation.

The project began in March 2019 when the indigenous communities living in the four proposed territories organized themselves internally to found the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara. It is from this starting point in the governance of the Guajajara indigenous peoples that it is possible to identify the beginnings of mobilization for their well-being and wellbeing, attracting other initiatives that share the common ideal of sustainable and social development.

Figure 1. Location of the KA'A PYAHU REDD+ project



Source: KA'A PYAHU REDD+ Project

With implementation of the activities for unplanned deforestation activities - AUDD, the KA'A PYAHU REDD+ project expects to reduce emissions by 200,000 tCO₂e per year.

1.2 Audit History

The project has not yet undergone any validation or verification process. However, the first validation period is expected to occur with the following dates:

Table 1. Audit history of the KA'A PYAHU REDD+ project

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	01-05-2021-30-04-2041	Verified Carbon Standard VCS		20

1.3 Sectoral Scope and Project Type

Table 2. Sectoral scope and type of the KA'A PYAHU REDD+ Project

Sectoral scope	Agriculture, Forestry, and Other Land Use
AFOLU project category¹	Reduction of emissions from deforestation and degradation (REDD+)
Project activity type	Avoiding unplanned Deforestation (AUDD)

1.4 Project Eligibility

1.4.1 General eligibility

The KA'A PYAHU REDD+ project is eligible under the scope of the VCS program for responding to REDD+ activities, under an independent project scheme for the validation and verification of GHG emission reductions in accordance with ISO 14064-2:2019 and ISO 14064-3: 2019. According with section 2.1 of the VCS Standard (V 4.7), the activities of the project are not excluded.

1.4.2 AFOLU project eligibility

According to the VCS Standard V 4.7, eligible REDD activities are those that reduce net GHG emissions by reducing deforestation and/or degradation of forests, which is the case of the KA'A PYAHU REDD+ project. Deforestation is the direct, human-induced conversion of forestland to non-forest land.

The analysis of land use categories in the project is based mainly on Mapbiomas² which utilizes a satellite imagery dataset used composed by Landsat 5 (Thematic Mapper), Landsat 7 (Enhanced Thematic Mapper Plus) and Landsat 8 (Operational Land Imager), Landsat sensors. The MapBiomass classification scheme is a hierarchical system with a combination of LULC classes compatible with the Food and Agriculture Organization (FAO) and IBGE classification systems (Souza, et al., 2020). In this regard, the project meets the internationally accepted definition of forest, and was qualified as forest for a minimum of 10 years before the project start date. The definition of forest may include mature forests, secondary forests, and degraded forests.

According with the VCS standard, activities covered under the REDD project category are those that are designed to stop planned (designated and sanctioned) deforestation or unplanned (unsanctioned) deforestation and/or degradation. In this context, as described in section 1.9, the actions carried out in the KA'A PYAHU REDD+ project, with the creation of the association, are oriented towards the defense, preservation and conservation of the environment. Specifically, by taking action to combat the environmental and social impacts caused by the construction of the power line. This project activity stops unplanned deforestation and reduces GHG emissions by stopping deforestation of forests.

1.4.3 AFOLU project eligibility

² <https://brasil.mapbiomas.org/atbd---entenda-cada-etapa/>

The project has not been registered under another GHG program.

1.5 Project Design

- ☐ Single location or installation
- ☒ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

Grouped Project Design

Not applicable as it is not a grouped project.

1.6 Project Proponent

The project proponents are the **indigenous association Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara**, which is responsible for the governance of the Guajajara people in the four territories and indigenous domains: Cana Brava Indigenous Land, Urucu-Juruá Indigenous Land, Rodeador Indigenous Dominial and Lagoa Comprida Indigenous Land, on which emission reduction actions are carried out. The Guajajara people have rights over land use and carbon rights. **Biofix Consultoria Ltda**, as an allied company that carries out the structuring of the project and manages the commercialization of VCU.

Table 3. Contact information

Organization name	Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara
Contact person	José Luciano Clemente Guajajara
Title	President
Address	Coquinho Village Area, S/N Countryside JENIPAPO DOS VIEIRAS - MA
Telephone	+ 55 99 98524-5222
Email	conselhosupremodecaciques@gmail.com
Organization name	BIOFIX CONSULTORIA LTDA.
Contact person	Cristiane Vancine Leme

Title	Director of operations Biofix Brasil
Address	Rua Cerqueira Cesar, Avenida Paulista 1636 CJ.4 PAV.15 SAO PAULO-SP
Telephone	+55 11 99522-0749
Email	crisleme@biofix.com.br

Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara, under its authority, appoints the company **BIOFIX CONSULTORIA LTDA** to prepare the project document, through a temporary association; with which it exclusively grants the rights to issue and sell the emission reduction certificates it generates, without prejudice to the fact that carbon rights will always belong to the indigenous council.

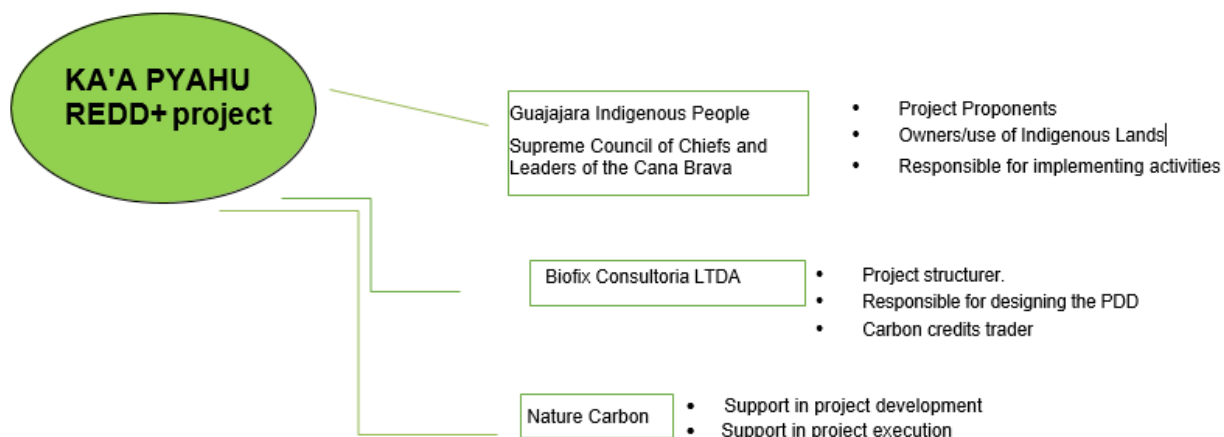
1.7 Other Entities Involved in the Project

The KA'A PYAHU REDD+ Project has an external participant, the company Nature Carbon, a partner of the Guajajara indigenous association, responsible for collaborating with the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara and with Biofix in the execution and development of the project.

Table 4. Other internal entities involved in the project

Organization name	Nature Carbon LTDA
Role in the project	Collaborate with the Association and Biofix to execute and develop the Project. Collect the requested documentation. Dialogue, interact and maintain a good relationship with ethnic authorities and administrative communities in the region. Witness and monitor socialization activities, socio-environmental workshops and Project actions in Indigenous communities. Inform about situations that impact the Project. When necessary, send the Minutes of recognition or possession of traditional authorities.
Contact person	Daniel Cavalcante
Title	Legal Representative
Address	Rua Urbano Santos, 155, room 402, Empress –MA.
Telephone	+55 99 8444-8181
Email	danielcavalcant.adv@gmail.com

Figure 2. Project organization chart



Source: KA'A PYAHU REDD+ Project

1.8 Ownership

The usufruct of the territories where the Ka'a Pyahu Project takes place belongs to the communities belonging to the Guajajara ethnic group, which are legally represented by the indigenous association Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara (Annex 011). The Cana Brava Indigenous Land, approved by DECREE No. 246, OF OCTOBER 29, 1991 (Annex 001); Urucu-Juruá Indigenous Land, approved by DECREE No. 382, OF DECEMBER 24, 1991 (Annexes 002), there was a rectification of the geographic coordinates of Decree No. 382, of December 24, 1991, which approved the administrative demarcation of the Indigenous Land Urucu/Juruá on FEBRUARY 10, 2003 (Annex 003); Lagoa Comprida Indigenous Land, DECREE OF FEBRUARY 10, 2003 (Annex 004) and Rodeador Indigenous Domain Area approved by DECREE N° 88,813, OF OCTOBER 4, 1983 (Annex 005), are under the possession of the people proposing the project, therefore owners of the carbon credits and of tradable carbon credits.

The Guajajara indigenous people who are part of the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara, making use not only of their rights of autonomy and self-determination, but also of the legal capacity that has been legally granted to them, which allows them to enter into contracts under the terms of civil and commercial law, through the signing of the Partnership and Cooperation Agreement (Annex 012), granted to BIOFIX CONSULTORIA LTDA, as the project's structuring entity, the express power to generate the issuance and commercialization of carbon credits (certified reductions in emissions GHG emissions) originating from the territories in which the Community Council exercises its jurisdiction.

The communities proposing the KA'A PYAHU REDD+ project are legally organized in a representative manner before the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara:

Table 5. Villages in the indigenous land of the project area

Indigenous lands	Villages represented	Attachment
T.I. CANA BRAVA- MEARIM REGION	39	Annex 06a
T.I. CANA BRAVA REGION SÃO PEDRO ISLAND AND CACIMBA VELHO	16	Annex 06b
T.I. CANA BRAVA RIO CORDA	52	Annex 06c
T.I. CANA BRAVA - MA 328	26	Annex 06d
T.I. CANA BRAVA - ARM OF CACHOEIRA COMPRIDA MEARIM	16	Annex 06e
T.I. CANA BRAVA BR - 226	73	Annex 06f
T.I. LAGOA COMPRIDA	17	Annex 07
T.I. RODEADOR	14	Annex 08
T.I. URUCU JURUÁ	14	Annex 09

Source: KA'A PYAHU REDD+ Project

According to updated data from March 2024 provided by SESAI (Special Secretariat for Indigenous Health), the Cana Brava Indigenous Land has an indigenous population of 11.528 people, has a health network of a DSEI (Special Indigenous Health District) and two base centers of health. The Urucu Juruá Indigenous Land in total covers a population of 994 people, with a DSEI and a Health Hub in its network. The Lagoa Comprida Indigenous Land in total covers a population of 727 people, with a DSEI and a Base Health Center, And Dominial Rodeador, 752 people, a base hub and a DSEI. The KA'A PYAHU REDD+ project benefits a total of 14.001 Guajajara indigenous people.

In turn, the indigenous association expressly stated that it did not hire another natural or legal person before signing the contract and committed itself not to hire another natural or legal person for the same purposes. Consequently, BIOFIX CONSULTORIA LTDA, as another project proponent, will be responsible for carrying out the necessary activities to achieve validation, verification, certification and commercialization of certified emission reduction units before the competent entities.

1.9 Project Start Date

The KA'A PYAHU REDD+ project begins on March 20, 2019, with the opening and founding of the indigenous association the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara (Annex 011). This activity is aligned with the governance line and is responsible for the legal representation of the Guajajara peoples of the four territories to access public policies, compensation, mitigations and the fight for the people's rights through legal means.

- Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara.

The association called Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara was founded on March 20, 2019 in Aldeia Coquinho, Cana Brava Indigenous Land, in the municipality of Jenipapo dos Vieiras in the state of Maranhão, northeast of Brazil. It is a non-profit civil association of indefinite duration that was founded to work with the Guajajara communities mentioned above and legally represent them in accordance with Brazilian legislation.

The objective of the association is, according to its statute, to develop actions to defend and contribute to improving the quality of life of the indigenous communities that are represented by the council, through: promotion of social assistance; promotion of culture and defense of historical and artistic communities. Defense, preservation and conservation of the environment, and promotion of sustainable development. Promotion of economic and social development and combating poverty in communities. Promotion of ethics and peace, respecting citizenship and human rights, in addition to democracy and others values present in the communities belonging to the Council.

To be able to carry out such actions, the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara is organized for its deliberations as follows: general assembly, deliberative council, board of directors, fiscal council, president and secretary. The form of internal organization occurs as follows, according to the statute: president, vice-president, secretary and treasurer (Annex 011). The deliberative council is made up of the president, treasurer, secretary and deputy secretary. The association has its own forms of application and intervention in accordance with collectively agreed principles.

The Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara, which represents the 267 villages that make up the KA'A PYAHU REDD+ project, has as one of its significant achievements, which is fundamental to this project, a prior consultation meeting with the communities of the Cana Brava, Lagoa Cumprida, Urucu Juruá and Rodeador indigenous communities on February 11, 2022 at their headquarters in the Coquinho village, in the municipality of Jenipapo dos Vieiras, with the purpose of formalizing the communities' interest in the structure, certify and commercialization of carbon credits project through an intermediation contract for these purposes (Annex 013 and 017).

The Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara, in accordance with the representation objectives mentioned above, promotes actions with the Brazilian state and competent bodies in order to ensure the right to health (Annexes 014 and 016) and the practice of citizenship and access to the registration documents necessary to access rights (Annex 015).

Furthermore, the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara has been fighting since its creation against the impacts in the environmental and social areas suffered by the construction of the energy line along the BR 226 highway in the Cana Brava Indigenous Land resulting from the Tucuruí Hydroelectric Plant, in the state of Pará, This is the first large-scale plant in the Amazon region, inaugurated at the end of 1984, and the transmission line towers for the interconnection of the North/Northeast System that runs through the Guajajara villages and causes social and

environmental harm in the region (Ribeiro Filho & de Campos, 2023). (Annexes 018, 018a, 018b, 018c, 018d, 018e, 018f, 018g, 018h, 018i and 018j).

Therefore, the activities carried out by the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara are aligned with the objective of the REDD+ Project, because: 1. It is carried out in the project area, 2. With people from the four Territories and Indigenous Domain, 3. Under the leadership of the Guajajara people in order to directly defend and ensure their rights by exercising indigenous citizenship under Brazilian legislation.

Complying with the start date requirements set out in the VCS standard for AFOLU projects, about the pipeline listing process (3.8.2 section), the start date of the project is set as follows:

Project start date	01-may-2021
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1.10 Project Crediting Period

Table 6. KA'A PYAHUU Project Crediting Period

Crediting period	20 years. According to the guidelines of the VCS Standard V4.7 the credit period of the project is 20 years.
Start and end date of first or fixed crediting period	01-May-2021 to 30-April-2041

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

- ☒ < 300,000 tCO₂e/year (project)
☐ ≥ 300,000 tCO₂e/year (large project)

Table 7. Estimated emissions reductions of the KA'A PYAHUU Project

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO ₂ e)
01-May-2021 to 31-December-2021	133,333.3
01-January-2022 to 31-December-2022	200,000
01-January-2023 to 31-December-2023	200,000
01-January-2024 to 31-December-2024	200,000
01-January-2025 to 31-December-2025	200,000
01-January-2026 to 31-December-2026	200,000

01-January-2027 to 31-December-2027	200,000
01-January-2028 to 31-December-2028	200,000
01-January-2029 to 31-December-2029	200,000
01-January-2030 to 31-December-2030	200,000
01-January-2031 to 31-December-2031	200,000
01-January-2032 to 31-December-2032	200,000
01-January-2033 to 31-December-2033	200,000
01-January-2034 to 31-December-2034	200,000
01-January-2035 to 31-December-2035	200,000
01-January-2036 to 31-December-2036	200,000
01-January-2037 to 31-December-2037	200,000
01-January-2038 to 31-December-2038	200,000
01-January-2039 to 31-December-2039	200,000
01-January-2040 to 31-December-2040	200,000
01-January-2041 to 31-April-2041	66,666.7
Total estimated ERRs during the first or fixed crediting period	4,000,000
Total number of years	20 years
Average annual ERRs	200,000

1.12 Description of the Project Activity

The forests in the project area are important areas for conservation. Although they are strategic ecosystems in the provision of environmental services, they are at risk of deforestation and invasion due to the exploitation of wood from local species, in addition to the risk of burning.

To promote the preservation of the four territories and domains and consequently support local cultures, actions must be generated that inject resources and means to protect the diversity of local fauna and flora in addition to the sustainable use of natural resources together with the strategic management of ecosystems.

It is essential that the risks of forest fires caused by human action are observed throughout the development of the project through strong monitoring in partnership with local fire fighters, in this case, Prevfogo.

REDD+ principles require that anthropogenic activities of deforestation and forest degradation are monitored and combatted strategically with the Guajajara people. In this way, actions were designed and coordinated with communities to reduce the aforementioned threats to forests in the project area and mitigate the impacts associated with GHG emissions.

The KA'A PYAHU REDD+ Project aims to formulate and promote local solutions for the development of a sustainable, participatory project to ensure environmental sustainability and mitigate the threats posed by the main agents and drivers of deforestation and forest degradation.

Through the implementation of actions in the four REDD+ investment lines: 1. Governance, 2. Health/Education/Infrastructure, 3. Monitoring and Reforestation and 4. Green Business, the aim is to avoid threats to local species through the implementation of a project REDD+ Reduction of Emissions due to deforestation and Forest Degradation, conservation of carbon reserves, sustainable management of forests and improvement of forest reserves in developing countries), providing payments for compensation for ecosystem services provided by the Guajajara people in the three territories and dominion.

Thus, the collective territories linked to the project seek, through their territorial autonomy, to carry out conservation processes and consequent valorization of their cultural identities and to promote the protection of biodiversity and the environmental wealth of their territories. Characteristics directly linked to the purpose of REDD+ projects, the desired sustainable local development, as well as the purpose of reducing emissions generated by deforestation.

The Project Proponent is distributing the finances, providing support for social development activities and implementing the monitoring program for the activities that will oversee BIOFIX BRASIL. The project activities, with which it is intended to comply with the resource plan and the development of the proposed objective, can be seen in the following table.

Table 8. KA'A PYAHU REDD+ lines of action.

Activity	Objective	Sub-activities	Indicator	Measuring Period	Related Stakeholders
CROSS LINES OF ACTION					
Institutional governance and strengthening	Apply training, provision, and authority strengthening actions as well as improve their culture and traditions.	Training workshops for leaders and communities in REDD+.	Indicators Number of people trained in REDD+.	Annual	State and Federal Universities Community Council Municipal Authorities Communities
		Territorial Monitoring – Internal Management Plan for territories.	Monitoring Plan produced by the community.		
		Acquisition of equipment to strengthen associations.	Purchase of Equipment.		
		Construction of cultural houses.	Number of houses built.		
		Support the participation of regional and national indigenous mobilizations (cultural exchanges).	Number of people who attended the event and minutes of participation.		
		Support for the movement of leaders for internal coordination in the territories.			

Participatory restoration, recovery, and/or rehabilitation of deforested and/or degraded ecosystems.	Conduct reforestation and native vegetation recovery in the riparian zone and degraded soils for conflicts of use with participation of the local communities.	Irrigation structure-distribution of water for the growth of crops.	Indicators	Annual	Community Council Municipal Authorities IBAMA ICMBIO FUNAI
		Planting of native species: buriti, tucumã, native seeds that support local crafts.	Number of structures built in communities. Number of native species planted.		
		Construction of nurseries for native species.	Number of nurseries of native species planted.		
		Training of local agents and forest guardians.	Number of trained agents.		
Monitoring and control plan of deforestation and forest degradation.	Monitor the changes due to deforestation in influence through satellite image processing and/or the ones taken from the project.	Territorial inspection.	Indicators	Monthly	BIOFIX will oversee the monitoring program through the joint work between expert trainers and keepers of the community.
		Preventing fires during critical periods.	Number of alerts reported per year.		
		Monitoring against harassment and threats of theft of internal natural resources.	Parameters for monitoring. Supports Satellite images		

			Images captured by drones. Images and videos captured in the field. Periodical monitoring reports		
Strengthening and encouragement of education	Encourage the enrollment in quality education and the completion of the study period by the students, guaranteeing access to higher education.	Construction, improvement and adaptation of the communities' school structure (Aldeia Flor reference). Construction of a computer laboratory Construction of sports courts for physical education and strengthening cultural sports activities (archery), Strengthening the mother tongue, training and internal training of education professionals in	Indicators Observation of built and improved school structures. Number of students benefiting. Acquisition of specific equipment. Number of trained professionals.	Annual	Municipal Authorities Community Council Universities MEC

		the mother tongue (support)	Number of workshops offered and the audience benefited.		
		Strengthening local culture in schools (crafts, traditional practices, indigenous school education – indigenous education) through workshops on traditional cultural activities with teachers and masters from the community),			
		Support for indigenous students in the city to continue their education,			
		Strengthening traditional festivals, such as: the girl's festival, the honey festival, the corn festival, the boys' festival, the cassava, potato, yam, pumpkin festival, and “Imitation”, moments of cultural transmission between generations through festivals and rituals.			

Prevention and health care program	Encourage health prevention and care of the community council settlers, guaranteeing a good quality of life.	Construction of hospital structure.	Indicators Evidence from the construction of health structures. Number of collections. Number of medicinal species planted. Number of joint efforts carried out with funding from the project. Number of trained agents. Material evidence of benefits in this type of mobility.	Annual	Municipal Authorities Community Council SESAI communes FUNAI
		Strengthening traditional medicine.			
		Collection of waste and garbage that remains in the territory and generates disease			
		Disease prevention through the planting of native species that are used in traditional medicine.			
		Support in the fight against pollution (collective effort).			
		Training in traditional medicine, support – conversation circles (studies in the area of pharmaceuticals, biology, partnerships in the manufacture of traditional medicines.			
		Purchase of vehicles for emergency/health trips.			

Green Business	Encouragement and support of income generation through sustainable activities.	Bee farming – beekeeping.	Number of trained creators and creations. Number of sales of crafts through activities supported by the project. Number of tanks and capabilities. Number of captives of native species.	Yearly	Communities Community Council and leaders ICMBIO IBAMA
		Native seed crafts.			
		Fish farming.			
		Native seedling bed and creation of native fruits, for example: mango, cashew nuts			
Infrastructure	Works and structures that facilitate and support local ways of life.	Drinking water treatment.	Number of beneficiaries and diseases avoided. Number of wells installed in communities. Formalization and number of initiatives created.	Yearly	Community Council Leadership FUNAI
		Installation of artesian well.			
		Creation of cooperatives to strengthen internal productive activities			
		Construction of a flour house.			

		Construction of houses/housing/housing structure.	Number of houses built.		
		Road construction.			
			Structures built and improvements generated.		

1.13 Project Location

The KA'A PYAHU REDD+ project is located in the state of Maranhão, in the northeast region of Brazil, under the jurisdiction of the Legal Amazon. Specifically, the project is located on indigenous lands of the Guajajara peoples: Cana Brava Indigenous Land, Urucu-Juruá Indigenous Land, Rodeador Indigenous Dominion Area and Lagoa Comprida Indigenous Land ([figure 1](#)). The total area of the project area is 178.483 hectares 165216,78 (this area was calculated using GIS software based on the coordinate system WGS_1984_UTM Zone 22S).

Table 9. Geometric centroid coordinates of the Project Area

Collective Territory	Extension (Ha) ³	Coordinates	
		Latitude	Longitude
Cana Brava/Guajajara	137.329,54	5° 37'6.02900"S	45° 32'43.01100 W
Lagoa Comprida	13.198,27	5° 19'28.17800"S	45° 44'5.278000 W
Rodeador	2.319,45	5° 42'12.19300"S	45° 13'42.49000 W
Urucu/Juruá	12.697,04	5° 22'30.76700"S	45° 55'0.432000 W

Source: SII FUNAI⁴

1.14 Conditions Prior to Project Initiation

1.14.1 Social parameters

1.14.1.1 Historic Contact-Guajajara

The Ka'a Pyahu project takes place in four territories and one indigenous domain, all with the Guajajara indigenous population in the northeast region of Brazil, state of Maranhão. The Guajajara began contact history around 1615 with French explorers. The self-styled Tenetehára people who were found along the Pindaré River went through periods of subjugation through Jesuit missions with a slavery bias around the 17th and 18th centuries. After the expulsion of the Jesuit colonies, the Guajajara freed themselves from the slavery system, but from the 19th century onwards, forced integration occurred through regional patronage systems (Schröder, 2021).

³ Declared area

⁴ http://sii.funai.gov.br/funai_sii/informacoes_indigenas/visao/visao_terras_indigenas.wsp

In the current Cana Brava Indigenous Land, which is part of the KA'A PYAHU REDD+ project, there was the biggest revolt ever known by the Guajajara at the end of the 19th century, where the chief Cauré Imana united a large number of villages and expelled the non-indigenous people in the current regions of Barra do Corda and Grajaú, the municipalities where the territories that are part of the project are located (Consiglio, 1997). This revolt was called the “Alto Alegre” revolt and is a historical milestone for the Guajajara, it ended with the invasion of the army and military policy that persecuted the Guajajara for years.

Since the last decades of the 20th century until then, the Guajajara people have suffered due to territorial invasions in which squatters threaten the security of the people within their lands. The threat of large projects such as the Grande Carajás Program and groups such as loggers also threaten their lands and the Ka'a Pyahu project through monitoring, reforestation and infrastructure activities is fundamental to supporting the security of the Guajajara people (Soares, 1983).

In addition to their history of contact with non-indigenous people, the Guajajara also had moments of hostility, such as with the Canela, Urubu-ka'apor and Guajá peoples.

The Guajajara, project proponents, have the right to possess and enjoy the territories that are registered in the KA'A PYAHU REDD+ project. They are: Cana Brava Indigenous Land, approved and registered in accordance with Decree N° 246, OF OCTOBER 29, 1991, the Lagoa Comprida Indigenous Land, approved and registered in accordance with Decree N° 313, OF OCTOBER 29, 1991, Dominial Rodeador, approved and registered in accordance with Decree N° 88,813, OF OCTOBER 4, 1983, and the Urucu Juruá Indigenous Land, approved and registered in accordance with Decree N° 382, OF DECEMBER 24, 1991. The four territories of the Guajajara Indigenous peoples are represented by the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara. The Guajajara people have usufruct and possession, the right to use local land and biodiversity, and the right to the carbon contained in these territories.

1.14.1.2 Population

The indigenous territories where the Guajajara people are located, represented by the Conselho Supremo de Caciques e Lideranças da Terra Indígena Cana Brava Guajajara, are mainly in three municipalities in the state of Maranhão: Barra do Corda, Grajaú and Jenipapo do Vieiras.

The Cana Brava indigenous territory is located in the Municipality of Barra do Corda and is 309 km from Imperatriz, capital of the State of Maranhão, with the geographic coordinates: Latitude: 5°33' 09.1"S, Longitude: 45°39' 13.0"W, and an altitude of 200 meters. The Lagoa Comprida Indigenous territory, located in the Municipality of Jenipapo dos Vieira, is 289 km from Imperatriz of the State of Maranhão, with the geographic coordinates: Latitude: 5°03 '28.0"S, Longitude 46°33' 26.7"W, and an altitude of 100 meters. The Urucu/Juruá territory, located in the Municipality of Grajaú, is 190 km from Imperatriz, State of Maranhão, with geographic coordinates: Latitude: 5°21'32.3"S Longitude, 45°55'10.9"W, and at an altitude of 100 meters. Finally, the Rodeador Territory, located in the Municipality of Barra do Corda,

is 337 km from Imperatriz, State of Maranhão, with the geographic coordinates: Latitude: 5° 43' 18.8"S, Longitude: 45° 13' 28.8"W, and the altitude of 200 meters ([figure 1](#)).

Barra do Corda

In the municipality of Barra do Corda are the Cana Brava and Rodeador indigenous territories. In this municipality, data from the last 2022 census, carried out by IBGE, the Brazilian Institute of Geography and Statistics⁵, indicates that a total of 5,335 people were identified who declared themselves indigenous, 53,270 brown people, 58 yellow people, 6,214 black people and 19,649 white people.

Table 10. Population of the municipality Barra do Corda.

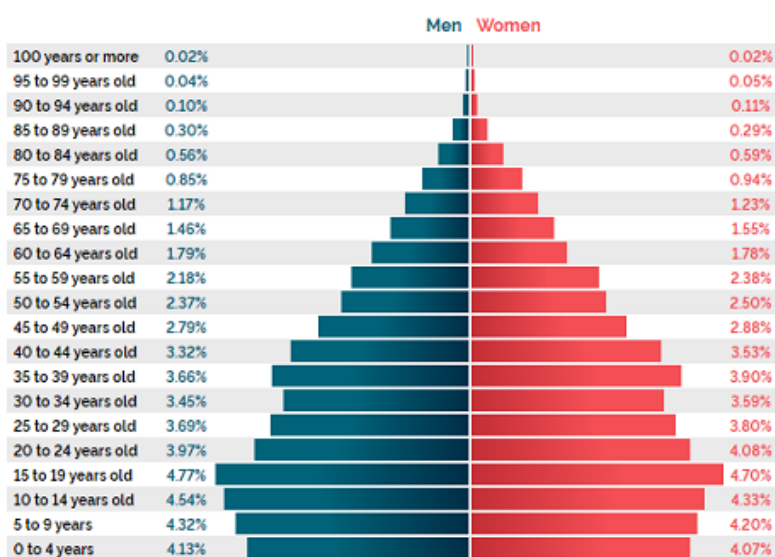
Population (2010)	Population (2022)	Color/Race	Municipality
20,325	19,649	White	Barra do Corda
4,577	6,214	Black	Barra do Corda
543	58	Yellow	Barra do Corda
53,953	53,270	Brown	Barra do Corda
3,432	5,335	Indigenous	Barra do Corda

Source: 2022 IBGE Demographic Census, adapted.

Of the total of 84,532 people living in the municipality of Barra do Corda, 5,552 are indigenous, representing 6.57% of the population. Of these, in indigenous territories 99% are indigenous and 1% non-indigenous. According to 2022 IBGE data, there are 4,500 people (81.05%) residing on Indigenous Land and 1,052 people (18.95%) outside indigenous territories. In the figure below is the age pyramid with the distribution of gender and age with the estimated lifespan of people in the municipality of Barra do Corda.

⁵ <https://cidades.ibge.gov.br/brasil>

Figure 3. Age distribution of the population in the Barra do Corda municipality.



Source: 2022 IBGE Demographic Census

Grajaú

The Urucu-Juruá Indigenous Land is located in the municipality of Grajaú, in the state of Maranhão. According to the most recent data from 2022, provided by IBGE, there are 7,646 indigenous people, which corresponds to 10.73% of the total population of the municipality. Since 2010 the growth in the rate of indigenous people was 3,510 people.

Table 11. Population of the municipality Grajaú.

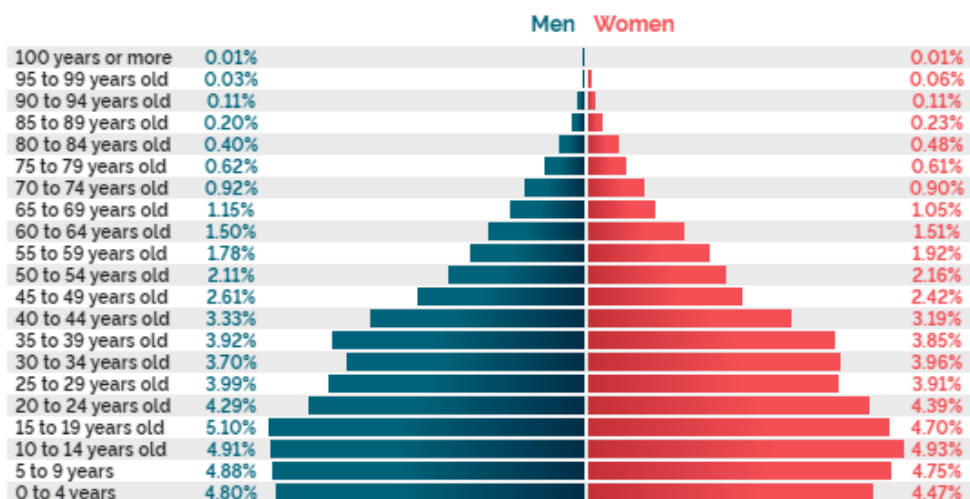
Population (2010)	Population (2022)	Color/Race	Municipality
15,369	15,459	White	Grajaú
4,483	6,336	Black	Grajaú
676	66	Yellow	Grajaú
37,430	44,365	Brown	Grajaú
4,135	7,646	Indigenous	Grajaú

Source: 2022 IBGE Demographic Census, adapted.

Of the region's total indigenous population, according to IBGE and 2022 data, 5,992 (75.59%) live within indigenous territory and 1,935 (24.41%) live outside indigenous territory. Within indigenous lands, 5,992 people (91.69%) are indigenous and 543 (8.31%) are non-indigenous.

As the age pyramid below indicates, the municipality's male population totals is 37,226 people (50.39%) and 36,646 females (49.61%).

Figure 4. Age distribution of the population in the Grajaú municipality.



Source: 2022 IBGE Demographic Census

Jenipapo das Vieiras

The Lagoa Comprida Indigenous Land, also a proponent of the project, is in the municipality of Jenipapo das Vieiras. Of the total 17,076 population residing in the municipality, 7,880 are indigenous, corresponding to 46.15% of the total population, with a growth of 44%.

Table 12. Population of the municipality Jenipapo das Vieiras.

Population (2010)	Population (2022)	Color/Race	Municipality
15,369	15,459	White	Grajaú
4,483	6,336	Black	Grajaú
676	66	Yellow	Grajaú
37,430	44,365	Brown	Grajaú
4,135	7,646	Indigenous	Grajaú

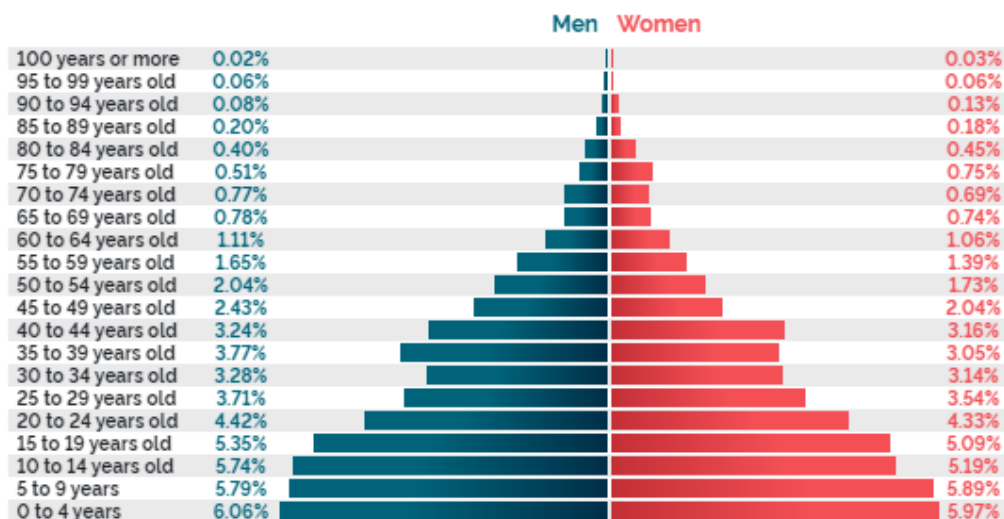
Source: 2022 IBGE Demographic census, adapted.

The indigenous populations residing in the demarcated territory correspond to 97.91%, with 7,701 people and residing outside the territory, 164 people, 2.09%. Within the indigenous territory, 97.3% are indigenous and 2.27% are non-indigenous.

The age pyramid of the municipality of Jenipapo das Vieiras, is made up of 8777 (51.40%) men and 8,299 women, which are equivalent to 48.60% of the total population, with an annual growth rate of 0.63 %.

In the three municipalities where the project proponents are located, there are no official records of the quilombola population.

Figure 5. Age distribution of the population in the Jenipapo das vieiras municipality.



Source: 2022 IBGE Demographic Census, adapted.

1.14.1.3 Education

Barra do Corda

In the municipality of Barra do Corda, where the Cana Brava and Rodeador indigenous territories are located, it is observed in relation to the educational profile, according to data from the last IBGE census in 2010 (IBGE, 2010), education chapter, the schooling rate for 6 to 14 year olds was of 95.8%. In comparison with other municipalities in the state, it was in position 152 out of 217. In comparison with municipalities across the country, it was in position 4,692 out of 5570. In relation to Basic Education Development Index (IDEB), in 2021, IDEB for the initial years of education fundamental level in the public school system was 4.6 and for the final years, 3.9. In comparison with other municipalities in the state, it was in positions 90 and 120 out of 217. In comparison with municipalities across the country, it was in positions 4,347 and 4,435 out of 5570.

Grajaú

In 2010, the school enrollment rate for 6 to 14 year olds was 96.3%. In comparison with other municipalities in the state, it was in position 132 out of 217. In comparison with municipalities across the country, it was in position 4,359 out of 5,570. In relation to IDEB, in 2021, IDEB for the initial years of education fundamental level in the public school system was 4.3 and for the final years, 3.8. In comparison with other municipalities in the state, it was in positions 143 and 137 out of 217. In comparison with municipalities across the country, it was in positions 4716 and 4524 out of 5570

Jenipapo dos Vieiras

In 2010, the schooling rate for 6 to 14 year olds was 80.7%. In comparison with other municipalities in the state, it was in position 217 out of 217. In comparison with municipalities across the country, it was in position 5547 out of 5570. In relation to IDEB, in 2021, IDEB for the initial years of education fundamental level in the public school system was 3.9 and for the final years, it was (no data). In comparison with other municipalities in the state, it was in positions 195 and (there is no data) 217. In comparison with municipalities across the country, it was in positions 4973 and (there is no data) 5570.

1.14.1.4 Health care

Table 13. Health indicators for the municipalities in the project area

Health indicator	Barra do Corda	Grajaú	Jenipapo dos Vieiras
Infant Mortality (2021). Deaths per thousand live births	16.68	20.88	33.99
Hospitalizations for diarrhea (2022) per 100 thousand inhabitants	2.4	60.9	146.4
SUS Health Establishments (2009). Establishments	42	29	7

Source: 2022 IBGE Demographic Census, adapted.

Barra do Corda

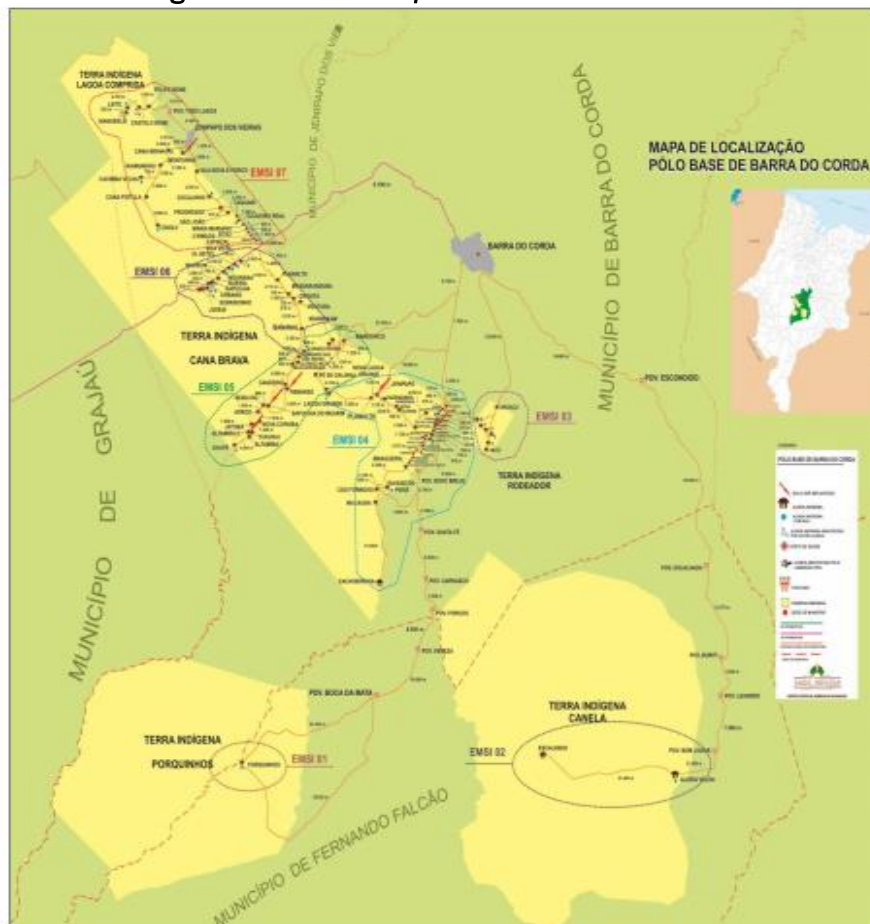
The average infant mortality rate in the city is 16.68 per 1,000 live births. Hospitalizations due to diarrhea are 2.4 per 1,000 inhabitants. Compared to all municipalities in the state, it is ranked 84 out of 217 and 147 out of 217, respectively. When compared to cities across Brazil, these positions are 1608 out of 5570 and 2388 out of 5570, respectively

Barra do Corda Base hub

The municipality of Barra do Corda has a Barra do Corda Base Center with specific health care for the indigenous populations of the region, it has seven Multidisciplinary Indigenous Health Teams-EMSI, distributed and operating in seven contiguous territories, as shown in the map.

This indigenous health hub serves the Lagoa Comprida, Rodeador and Cana Brava Indigenous Lands, proponents of the KA'A PYAHU REDD+ project. It covers the three municipalities registered in the project and covers a total of seven indigenous territories, 273 villages and a total population of 15,712 people according to data from the Special Secretariat for Indigenous Health (SESAI) from 2023.

Figure 6. Location map of Barra do Corda Base hub.



Source: Secretariat of Indigenous Health / Special Indigenous Health District

Grajaú

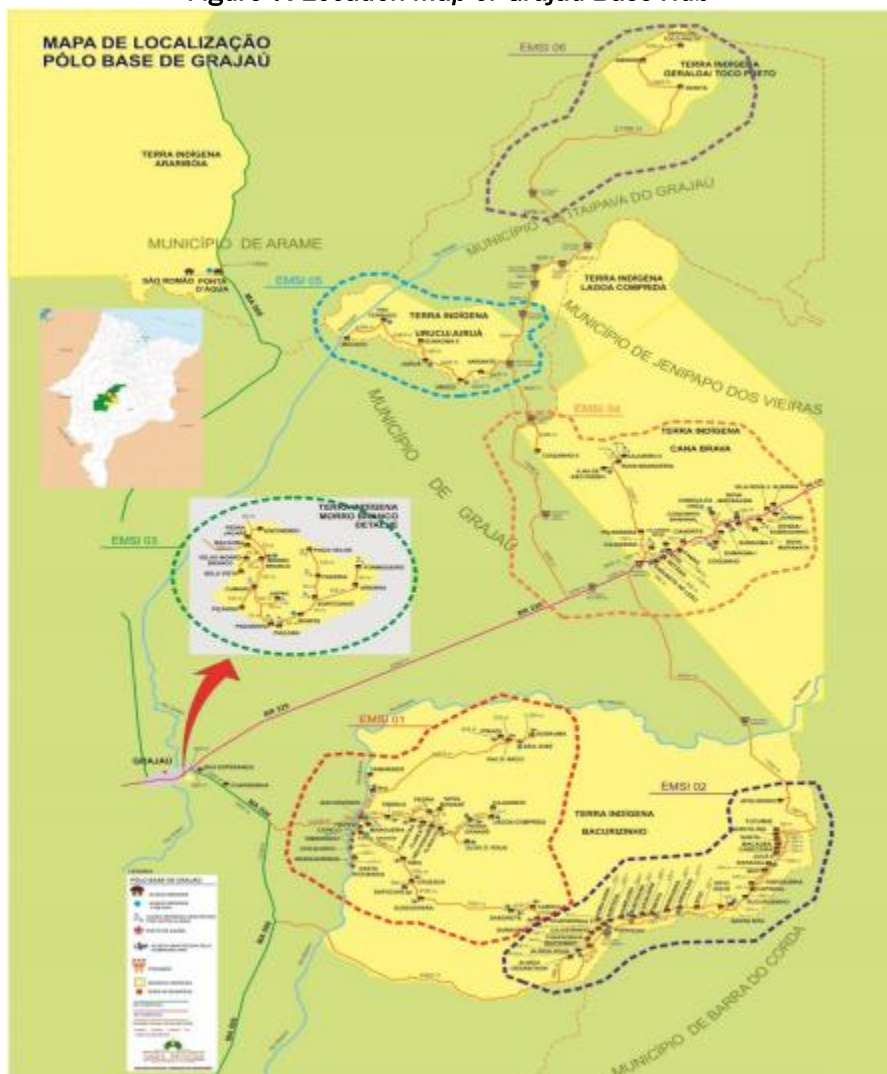
The average infant mortality rate in the city is 20.88 per 1,000 live births. Hospitalizations due to diarrhea are 60.9 per 1,000 inhabitants. Compared to all municipalities in the state, it is ranked 48th out of 217 and 103rd out of 217, respectively. When compared to cities across Brazil, these positions are 1,072 out of 5,570 and 909 out of 5,570, respectively.

Grajaú Base Hub:

The Grajaú Base Hub based in the municipality of Grajaú has six Multidisciplinary Indigenous Health Teams-EMSI, distributed and operating in six contiguous territories, as shown in the map.

The DSEI Grajaú covers six Indigenous Lands, among which three Indigenous Lands from the KA'A PYAHU REDD+ project are covered: Cana Brava, Lagoa Comprida and Grajaú. It covers a total population of 8,571 people and four municipalities.

Figure 7. Location map of Grajaú Base Hub



Source: Secretariat of Indigenous Health / Special Indigenous Health District

Jenipapo das vieiras

The average infant mortality rate in the city is 33.99 per 1,000 live births. Hospitalizations due to diarrhea are 146.4 per 1,000 inhabitants. Compared to all municipalities in the state, it ranks 9th out of 217 and 86th out of 217, respectively. When compared to cities across Brazil, these positions are 352 out of 5570 and 459 out of 5570, respectively.

1.14.1.5 Unmet Basic Needs

Barra do Corda

It presents 13.5% of households with adequate sanitation, 74% of urban households on public roads with trees and 3.4% of urban households on public roads with adequate urbanization (presence of drains, sidewalks, paving and curbs). When compared to other municipalities in the state, it is ranked 72nd out

of 217, 77th out of 217 and 54th out of 217, respectively. When compared to other cities in Brazil, its position is 4,090 out of 5570, 2,844 out of 5,570 and 3,771 out of 5570, respectively (IBGE, 2022)

Grajaú

It presents 4.8% of households with adequate sewage, 89.5% of urban households on public roads with trees and 4.5% of urban households on public roads with adequate urbanization (presence of drains, sidewalks, paving and curbs).). When compared to other municipalities in the state, it is ranked 136 out of 217, 24 out of 217 and 46 out of 217, respectively. When compared to other cities in Brazil, its position is 4823 out of 5570, 1544 out of 5570 and 3588 out of 5570, respectively (IBGE, 2022)

Jenipapo dos Vieiras

It presents 2.6% of households with adequate sewage, 95.4% of urban households on public roads with trees and 4.6% of urban households on public roads with adequate urbanization (presence of drains, sidewalks, paving and curbs). When compared to other municipalities in the state, it is ranked 173 out of 217, 11 out of 217 and 45 out of 217, respectively. When compared to other cities in Brazil, its position is 5108 out of 5570, 850 out of 5570 and 3563 out of 5570, respectively (IBGE, 2022).

1.14.1.6 Economic activities

The main subsistence activity of the Guajajara people, proponents of the KA'A PYAHU REDD+ project, is farming, especially the planting of cassava, corn, rice, pumpkin, watermelon, beans, broad beans, yams, yams, sesame, peanuts. In the dry season, from May to November, drilling, felling, burning, digging and cleaning are carried out, while from November to February planting and meadows are carried out.

According to (Schröder, 2021), the planted areas per residential unit vary between 1.25 ha and 3.55 ha per domestic unit, or between 0.25 ha and 0.71 ha per individual, respectively. This variation depends mainly on the involvement of communities and individuals in the marketing of agricultural products. Fishing is mostly practiced in riverside villages. The Guajajara usually fish around 36 different species, with the yam, cascudo, lamprey, mandi, pacu, piau and traíra being the most common. In recent years, however, in several community projects, small dams have been built near some villages that are far from rivers. For the residents of these villages, the dams allow both subsistence and commercial fishing.

The Guajajara traditionally hunt more than fifty-six species, the most common being peccary, agouti, jacamim, jacu, peccary and several species of monkeys and armadillos. In a part of the Guajajara lands, hunting became more productive again during the 1990s after more efficient controls over land limits were initiated by the Indians themselves.

Collecting is still practiced by almost all Guajajara. Collection activities, however, are increasingly being replaced by fruit growing in villages and farms. Currently the Guajajara plant around thirty types of fruit trees and palm trees. The only forest product still collected in larger quantities for commercial purposes is honey.

Economic relations with non-indigenous people are based on both material and monetary exchanges. The most common sources of income are the sale of agricultural products, the sale of handicrafts and

temporary work (for the settlers) or permanent work (for Funai). A very serious problem is the predatory commercialization of the areas' natural resources through concessions to loggers and hunters, in order to obtain small profits in the short term to, for example, buy medicines not provided by deficient government services.

1.14.1.7 Social and political organization

According to (Schröder, 2021) Guajajara villages are long, round or quadrangular. They are preferably located on the banks of rivers or, in the absence of watercourses, near lakes in the forest. The proximity to a road can be another attractive factor, for selling handicrafts, for example.

The villages, formerly very small and of temporary existence, are now permanent and rarely transferred. They can be made up of a single family, but in some cases they can have up to four hundred or more residents. The houses, built in the regional peasant style, are generally inhabited by nuclear families. Villages usually maintain their independence and rarely form regional coalitions, but there are diverse kinship, marriage and ritual relationships between communities.

In kinship, the most important unit is the extended family, which is made up of a number of matrilineal nuclear families linked together by kinship ties. It is, in essence, a group of related women under the leadership of a man. There are no moieties, clans or lineages, nor any right or obligation that is transmitted through a specific line of descent.

Residence after marriage is with the woman's parents, the man's in-laws, (uxorilocality). Many heads of extended families try to keep as many women as possible, even adopting the daughters of deceased men they used to call "brothers". They try to arrange marriages for these girls to obtain sons-in-law, who must live at least one or two years with their in-laws, providing various types of services. If the head of the family has enough prestige, he manages to get his sons-in-law to permanently settle with him, thus increasing the number of collaborators and attracting co-partisans to form a faction in the village.

The main traditional criteria for assuming leadership involving kinship politics (individual qualities and a base of co-partisans through consanguinity and affinity) have become less important, compared to the demands of knowing how to deal with the surrounding society. This concerns, firstly, the ability to relate to government bodies and have a political stance in relation to the local community, and individual qualities (knowledge of Portuguese and diplomatic talent, among others).

Each village has its own chief, but there are villages with more than one because of rivalries between several extended families. Some chiefs try to extend their influence to neighboring villages, but their authority is very unstable and can be challenged at any time by competitors from their own village. In this game for power, the indigenous body usually intervenes to promote its own protégés, who may be weak characters, with no true base in the villages.

1.14.1.8 Cosmology

Guajajara cosmology is typical of the Tupi-Guarani peoples, distinguishing four categories of supernatural

beings, which receive the generic designation of karowara: (1) the creators or cultural heroes, responsible for the creation and transformation of the world, being Maíra and the twins Maíra-ira and Mucura-ira the most important and Zurupari, the creator of pests and insects, venomous snakes and spiders, a much feared cultural hero; (2) the "owners" of the forests (Ka'a'zar), the waters (Y'zar), the hunts (Miar'i'zar) and the trees (Wira'zar), who are hostile and greatly feared by its evil power; (3) the azang, wandering spirits of the dead, also greatly feared; and (4) the piwara, animal spirits. Many Guajajara no longer believe in these beings, because of missionary activities (Wagley, 1973).

The mythology is a mix of Tupi, European and African motifs. There are three main categories of myths: (1) cultural hero myths; (2) myths that point to a moral; and (3) animal myths. In all the myths recorded so far, the role of Maíra stands out. A very important myth to explain the world from the Guajajara point of view is that of the twins Maíra-ira and Mucura-ira.

The major traditional rituals include the important Honey Festival, held in September or October, during the dry season, and which requires several months to prepare. It played a very important role in good relations between villages. The Corn Festival (awashire-wehuhau), also called the "shaman festival", takes place every year during the rainy season, during the growing season of this vegetable. Its purpose was to ensure a good harvest and protect the corn against the actions of the Azang. Therefore, its main characteristic was the pajelança. The Moqueado rite, held on the same occasion as part of the Corn Festival, marks the end of puberty for the participating teenagers. Moqueado is still practiced at irregular intervals, but it has become somewhat profane, often leaving only the culinary part of the rite to accompany political meetings.

A person's life cycle is still usually accompanied by a series of rituals. Among these, initiation rituals, particularly those of girls, are the most eye-catching and rich in meaning. Additionally, there are a series of rituals to ask permission from Maíra to plant, from Miar'i'zar to hunt and from Y'zar to fish.

Shamanism

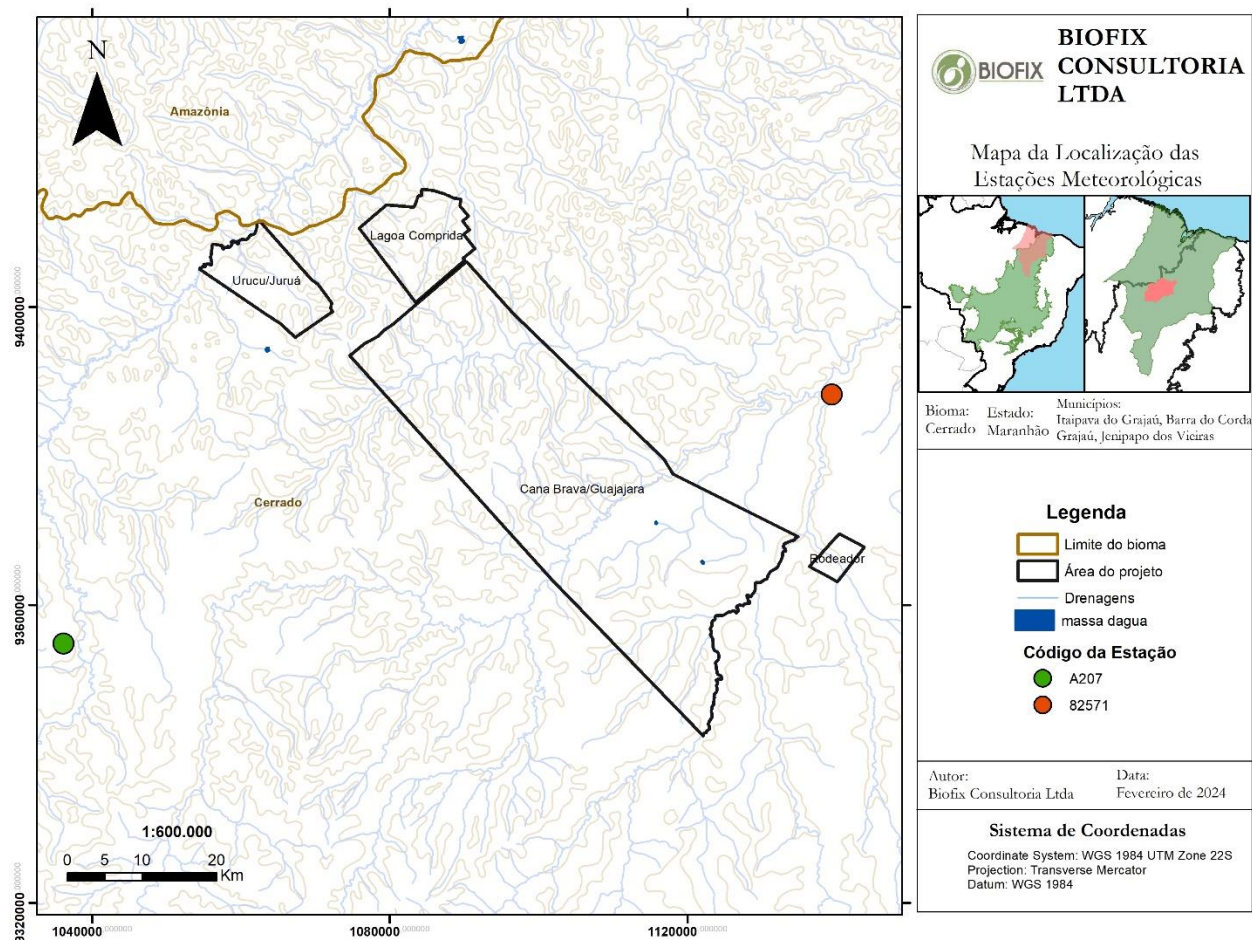
In ancient times, most Guajajara men tried, at any cost, to be shamans, but few were successful and gained fame. The power and reputation of the shamans depended on the number of supernatural beings they knew how to "call". Highly recognized shamans could also become powerful leaders. Pajelança is an almost exclusively male activity. The main function of the shamans is still to heal and celebrate the festivals of Maíra and the "allowance", a ritual of offerings in favor of sick people. Shamanism is often seen as ambiguous, because shamans' powers can be used for both positive and negative goals.

1.14.2 Biophysical Parameters

1.14.2.1 Climatology

To describe the climatology of the project area, particularly the variables of precipitation, temperature, and relative humidity, data from two meteorological stations in the National Institute of Meteorology's⁶ database were consulted. The two stations were selected because they are the closest stations to the project area, however, station Barra do Corda is the closest and less than 20 kilometers away. For the analysis of the two stations, data were consulted from the year 2000 (82571 station) and 2008 (A207 station), from which time they are available.

Figure 8. Location of the meteorological stations.



Source: KA'A PYAHU REDD+ Project.

⁶ <https://portal.inmet.gov.br/>

Table 14. Data from meteorological stations

Station	Name	Type	Municipality	Latitude	Longitude	Altitude
82571	Barra do Corda	Conventional	Barra do Corda	-5.506666	-45.236944	154.19
A207	Grajaú	Automatic	Grajaú	-5.816111	-46.162222	232.29

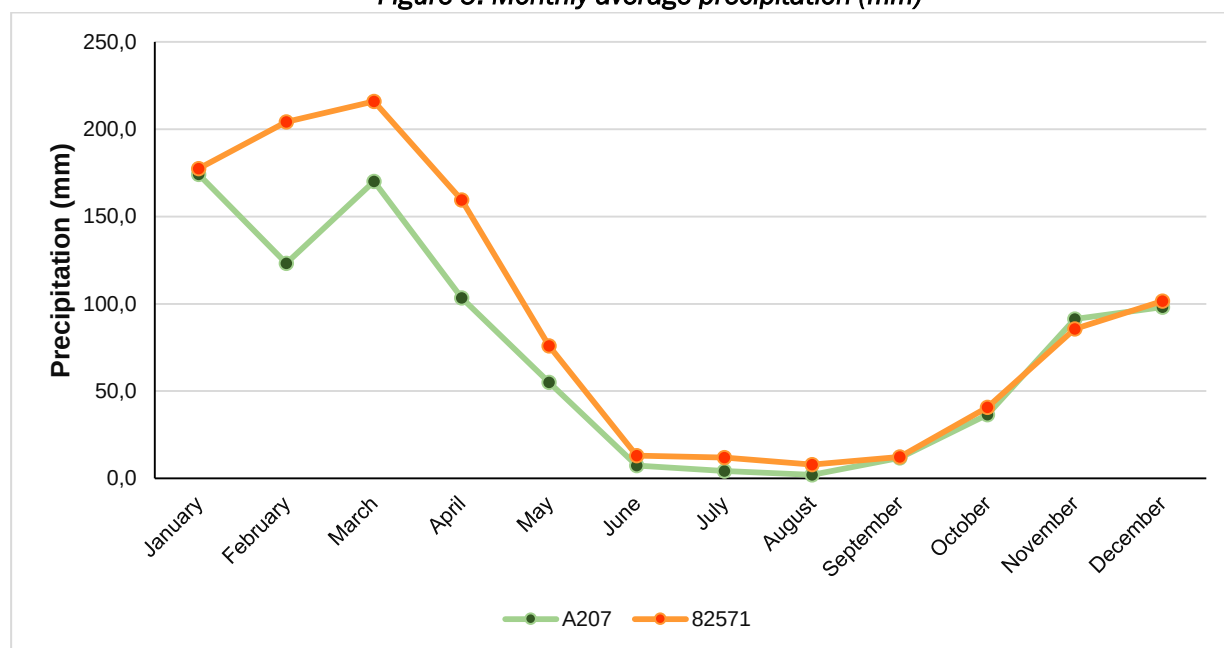
Source: National Meteorology Institute INMET.

Precipitation

Rainfall in the project area is characterized by a long period of rain, starting in October and extending through May, with January and March being the months with the highest rainfall, with maximum values that can reach up to 215 millimeters, and an average of 17 days of rain.

Following this, between June and September there is a dry period, with monthly average rainfall of less than 10 mm and no rainy days.

According to the data from the stations, the total annual precipitation varies between 878 mm (A207 Station) and 1105 mm (82571 Station), with 991 mm being the average between the two stations. The total number of annual days varies between 101 and 133, respectively.

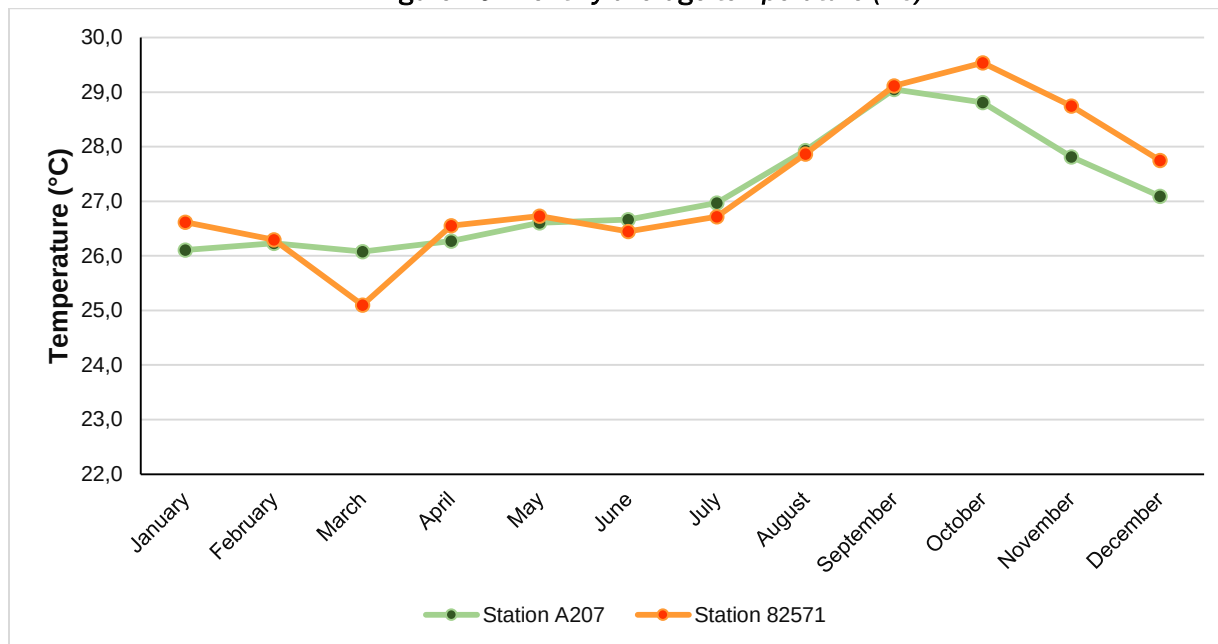
Figure 9. Monthly average precipitation (mm)


Source: Data from National Meteorology Institute INMET, adapted.

Temperature

According to data from the two automatic stations, the average annual temperature is 27.2 °C, with values varying between 25.6 °C and 29.3 °C.

Figure 10. Monthly average temperature (°C)



Source: Data from National Meteorology Institute INMET, adapted.

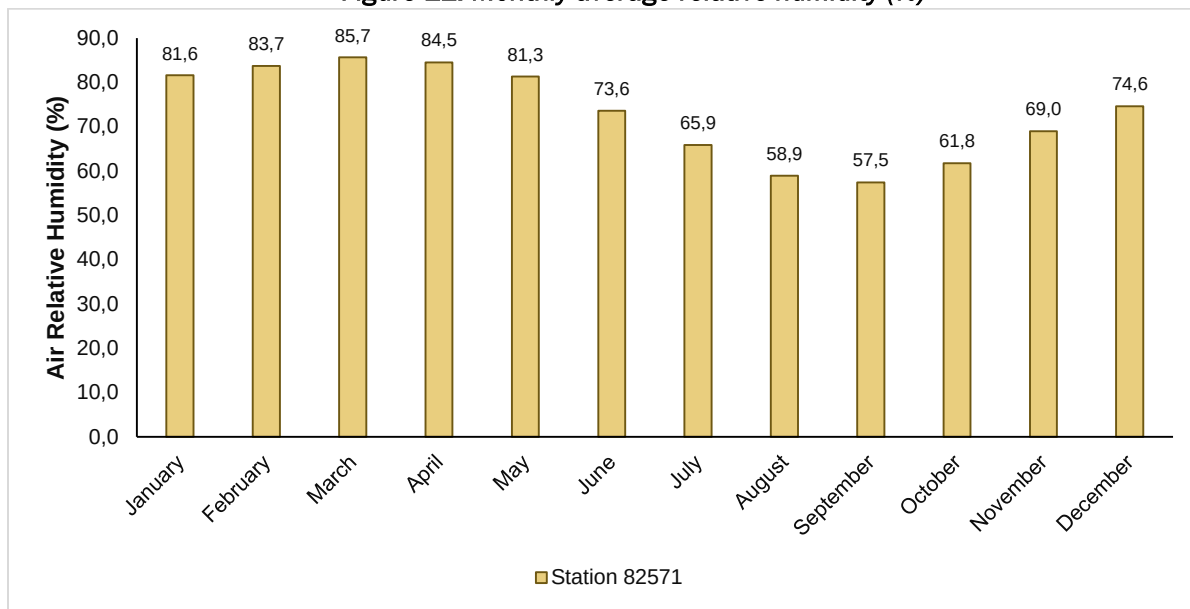
Generally, the temperature remains constant throughout the year. However, there is a difference of approximately 3.2 °C between the hottest months (September and October) and the coldest months (February and March).

Air Relative Humidity

For the air relative humidity variable, only the conventional station was taken into consideration, as it was the only that presented data for this variable.

The annual average relative humidity in the project area is approximately 73.2%. According to the station data, there is a correlation between precipitation and relative humidity. That is, the months with the highest precipitation (January through April) correspond to the highest humidity values, reaching values of up to 85% relative humidity. Conversely, the driest months (July to September) have the lowest relative humidity values, with values of up to 65% relative humidity. The difference between the months with the highest and lowest humidity is approximately 30%.

Figure 11. Monthly average relative humidity (%)



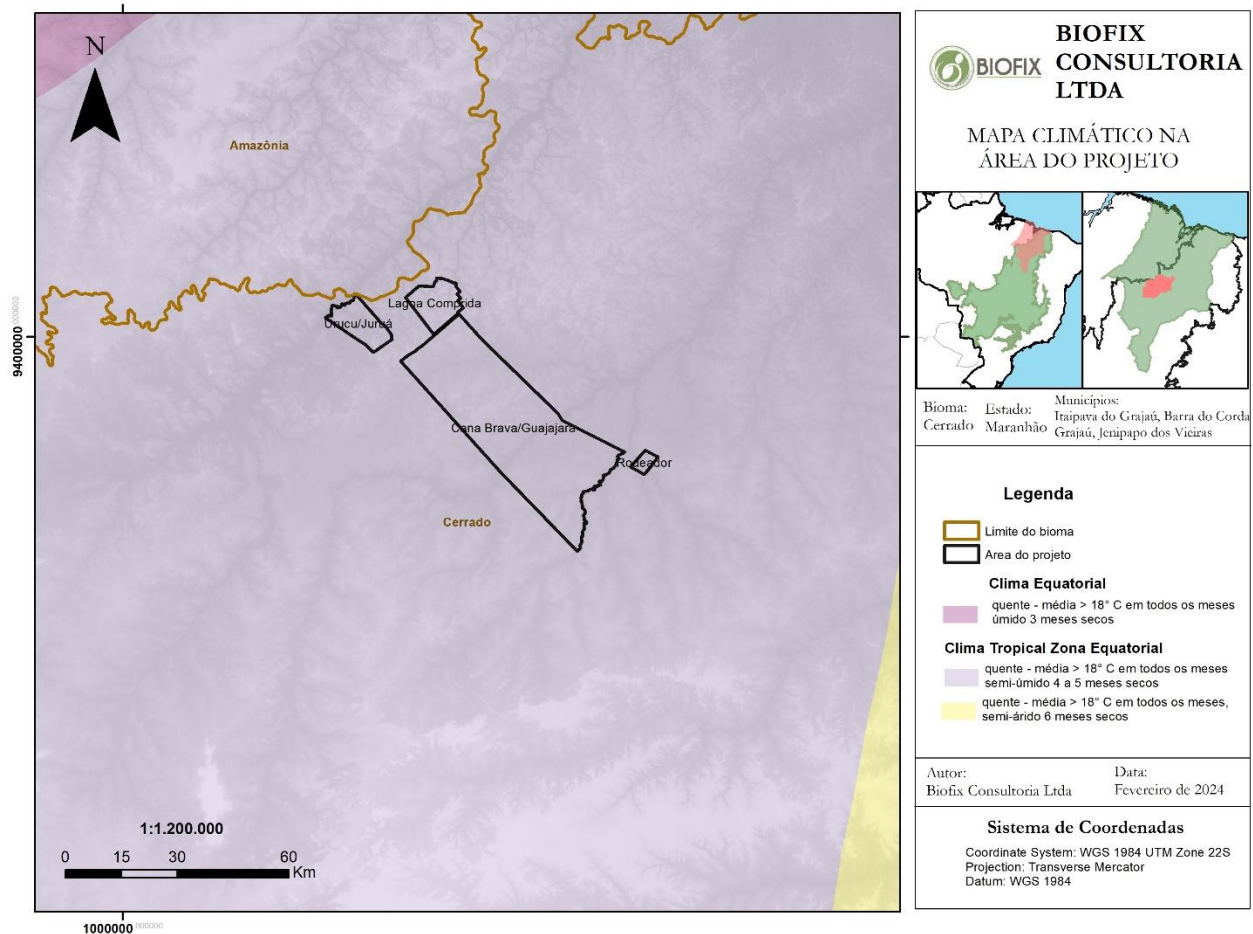
Source: Data from National Meteorology Institute INMET, adapted.

Climate Classification

According to the climate map of Brazil at a scale of 1:500.000 (Instituto Brasileiro de Geografia e Estatística IBGE, 2002), the project area is characterized by a humid equatorial climate, marked by the occurrence of 3 dry months and an average temperature above 18°C during all months of the year.

On the other hand, according to the Koppen Climate classification for Brazil, the project area has an **Aw** climate (Tropical with dry winter). The letter A indicates the climatic zone, defined by the temperature and precipitation, which in this case is *tropical*, with an annual average temperature above 18°C. The letter w considers the distribution of precipitation, in this case, it means with a dry winter, where the precipitation of the driest month is less than 25 mm (Alvares, Stape, Sentelhas, Gonçalves, & Sparovek, 2013).

Figure 12. Climate classification of the project area



Source: IBGE (2002), adapted.

1.14.2.2 Hydrography

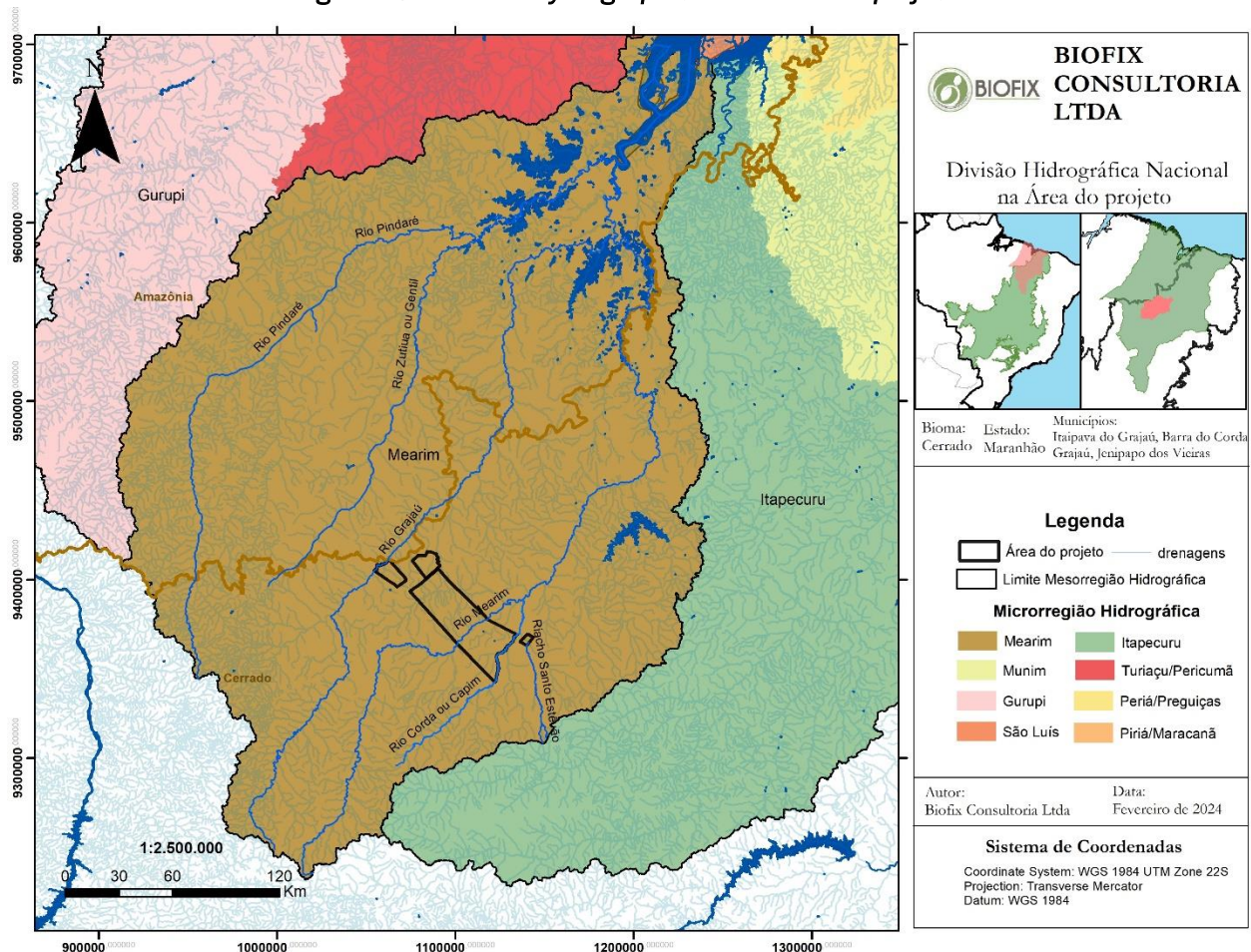
National Hydrographic Division DHN

The National Hydrographic Division of Brazil at a scale 1:250.000 was updated in 2021 in a joint effort by the Brazilian Institute of Geography And Statistics (IBGE) and the National Water Agency ANA (IBGE. Coordenação de Recursos Naturais e Estudos Ambientais, 2021). This division comprises three levels: The first level is the Hydrographic Macroregion (MacroRH), which divides Brazil into 12 hydrographic regions established in Resolution No. 32 of 15.10.2003 (MINISTÉRIO DO MEIO AMBIENTE. CONSELHO NACIONAL DE RECURSOS HÍDRICOS, 2003). The project area is located in the Atlântico Nordeste Ocidental macroregion, which has an extension of 275.000 km² and it is a region of great economic importance for the Northeast economy, since it has significant water potential for the development of agricultural activities.

The second level divides the country into 54 hydrographic mesoregions (MesoRH), based on hydrographic units that consider land use and resource management factor. The Nordeste Ocidental macroregion comprises 3 mesoregions, Gurupi, Itapecuru, and Mearim, being located the project area is located in

the Mearim mesoregion, which has an extension of 101.998 km² and whose main rivers are Mearim river, Grajaú river, Pindaré river, and Zutiua river (Algarves, Pereira, Fernandes, Alves, & Machado, 2016).

Figure 13. National hydrographic division in the project area



Source: adapted from (IBGE. Coordenação de Recursos Naturais e Estudos Ambientais, 2021).

The third level comprises 302 hydrographic microregions micro RH, based on water planning units. The Mearim mesoregion comprises 8 microregions, being the project located in the Mearim microregion. This means that the project area is located in the same meso- and microregion. In general terms, the National Hydrographic Division (DHN) considers economic, land use, and resource management factors.

Hydrographic basins in Brazil.

In contrast to the National Hydrographic Division, the Brazilian river basin division is based on the concept of a river basin being "an area of land surface bounded by watersheds that collects and conveys, through slopes, rivers and streams, water from precipitation to an outfall, a single point of exit, located at a lower point in the relief".

The different basins according to the levels of the project area are shown below:

Table 15. Hydrographic basins in the project area

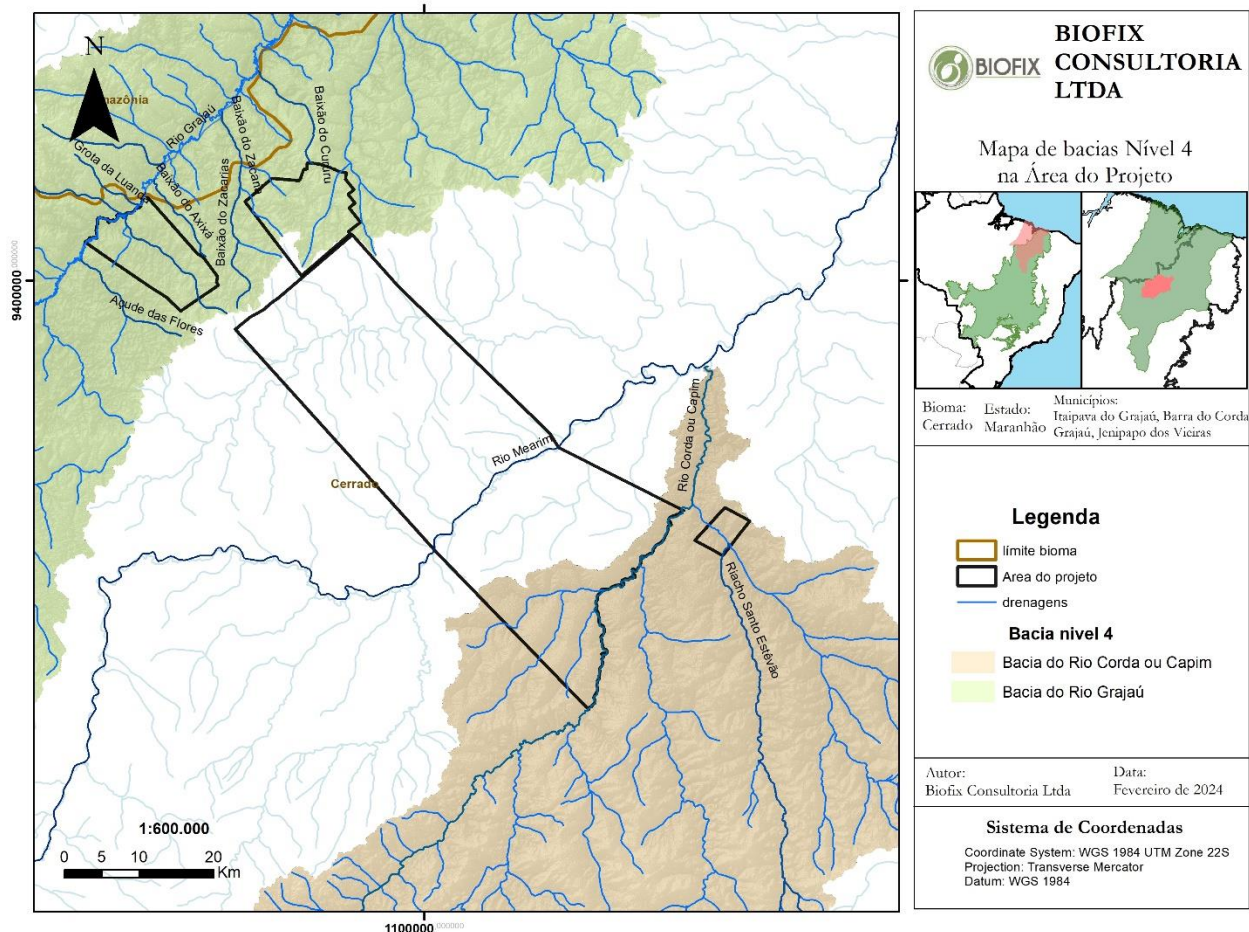
Level	Basin Name	Main River	Main tributaries
3	Rio Mearim	Mearim river	Pindaré, Grajaú, Zutiua, Santana, Piratininga, Ipixuna, Enjeitado, das flores, Corda.
4	Rio Corda	Corda river	Pau Grosso, Riacho Fundo, Ourives , Baixão do Sabonete, Baixão Pau Ferrado, Baixão do Zacarias, Baixão do Cururu.
	Rio Grajaú	Grajaú river	Santana, Riacho Santo Estêvão, Bacia do Baixão do Limão, Igarapé do Salgado.
5	Lagoa Comprida	Baixão da Lagoa Comprida	
	Capim Duro	Baixão do Capim Duro	Baixão do Fundo, Baixão do Francisco, Baixão do Crioli
	Baixão do Sabonete	Baixão do Sabonete	
	Rio Ourives	Ourives river	Brejo do Curicaca, Riacho Santo Estêvão.
6	Baixão do Crioli	Baixão do Crioli	
	Baixão do Zacaria	Baixão do Zacaria	
	Barro Branco-Fogo	Baixão do Barro Branco e Baixão do Fogo	
	Santo Estêvão	Riacho Santo Estêvão	

Data from (IBGE. Coordenação de Recursos Naturais e Estudos Ambientais, 2021).

The KA'A Y PYAHU project as mentioned in the Mearim river basin of the third level, as a mentioned before, the main rivers are Pindaré, Grajaú, Zutiua and Corda. The Mearim river originates on the northern slopes of Serra da Menina, in the extreme south of the Municipality of Grajaú, flowing into the Atlantic Ocean, 24 km after the confluence with the Pindaré river (Cunha, 2003).

There are two level 4 basins in the project area: The Grajaú river basin, which borders the Urucuru indigenous territory to the north. Of the main tributaries of the Grajaú River, the ones that pass through the project area are Baixão do Zacarias, Baixão do cururu, and Baixão do Axixá. The basin of the Capim river, which has as its main river the Capim or Corda river, which is the limit on the Cana Brava indigenous territory. One of the main tributaries of the Capim River is the Santo Estêvão stream, which runs through the Rodeador indigenous territory.

Figure 14. Level 4 basins in the project area

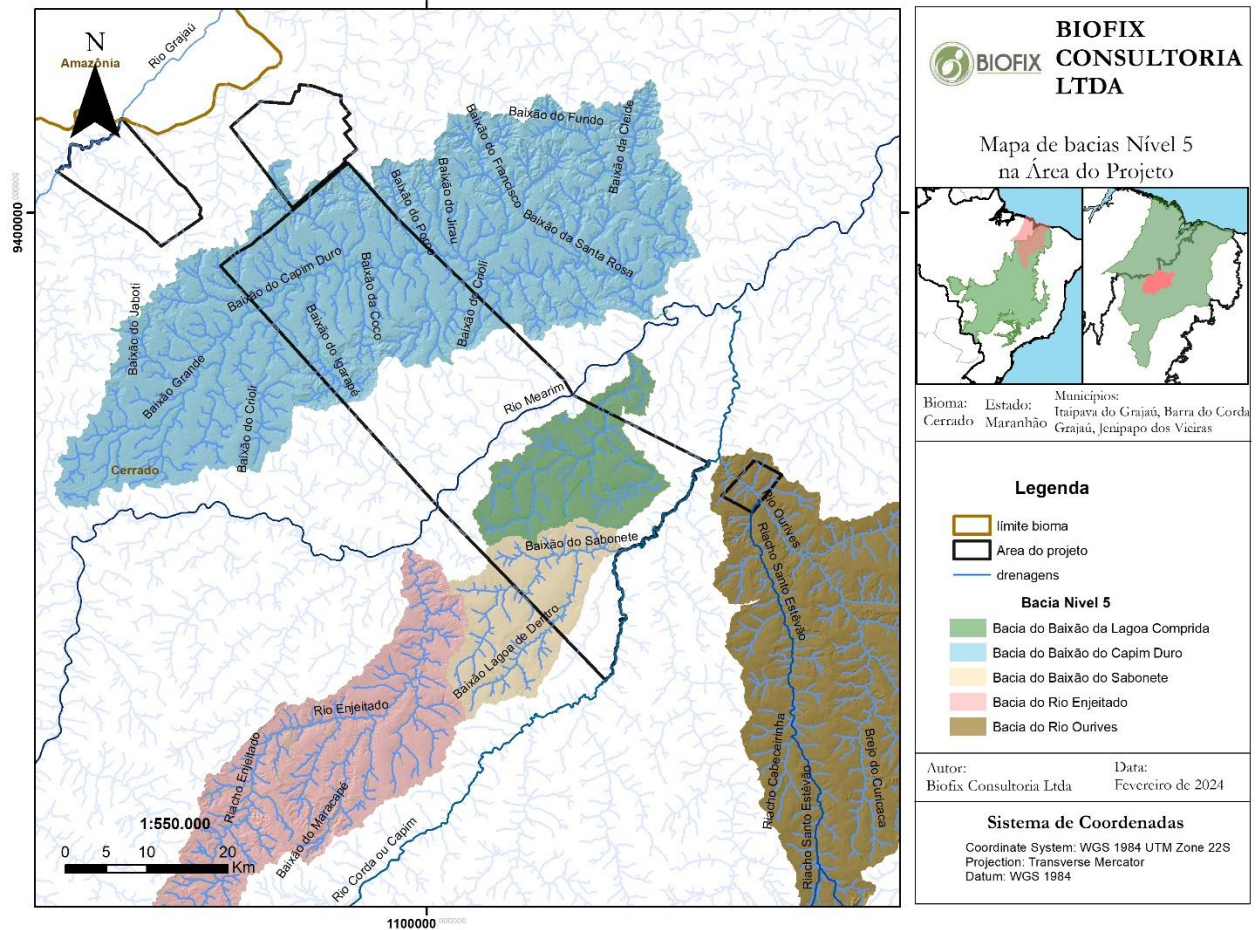


Source: adapted from (IBGE. Coordenação de Recursos Naturais e Estudos Ambientais, 2021).

Regarding type 5 level basins, the project area partially includes 4 basins: the Ourives river basin and the Sabonete basin, which both flow into the Capim river. The others two, the Lagao comprida basin and Capim duro basin flow into the Mearim river. With the exception of the Sabonete river basin, the other three basins cross the indigenous territory of Cana Brava. The Capim duro basins has as its main tributaries the Baixão do Francisco, Baixão do Crioli, and the Baixão do Coco, this which originates within the limits of the project area.

Similarly the Sabonete basin has the sabonete river as its main watercourse, and its originates within the limits of the project area, especially in the indigenous territory of Cana Brava. This demonstrates the importance of the project from a hydrographic point of view, since there are basins that feed important rivers such as Mearim river and Capim river.

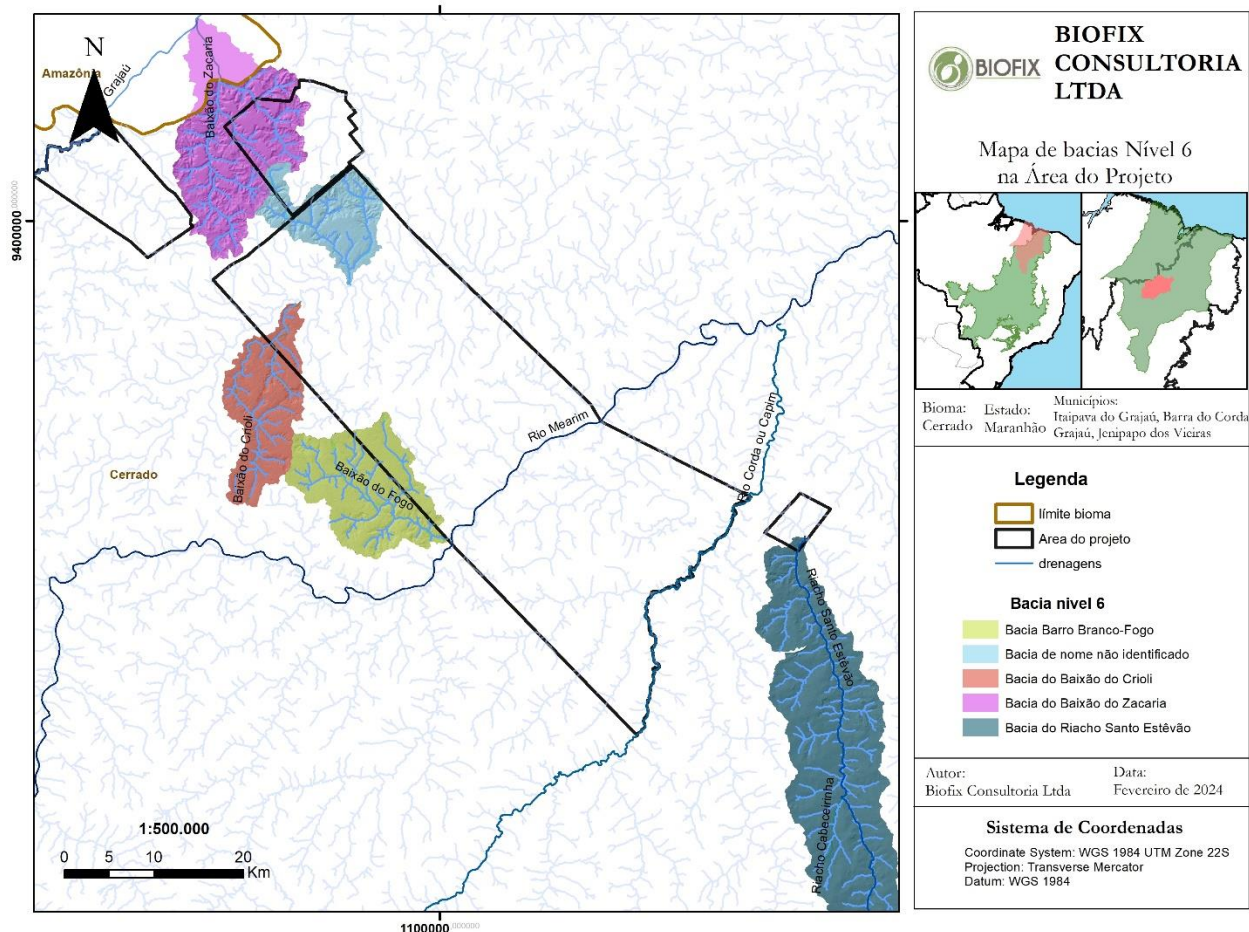
Figure 15. Level 5 basins in the project area



Source: adapted from (IBGE. Coordenação de Recursos Naturais e Estudos Ambientais, 2021).

Finally, the limits of the project area overlap superficially with 4 hydrographic basins of level 6. Of this level, the Zacaria River basin stands out especially, within the Lagoa Comprida indigenous territory, which flows into the Grajaú River.

Figure 16. Level 6 basins in the project area.



Source: adapted from (IBGE. Coordenação de Recursos Naturais e Estudos Ambientais, 2021).

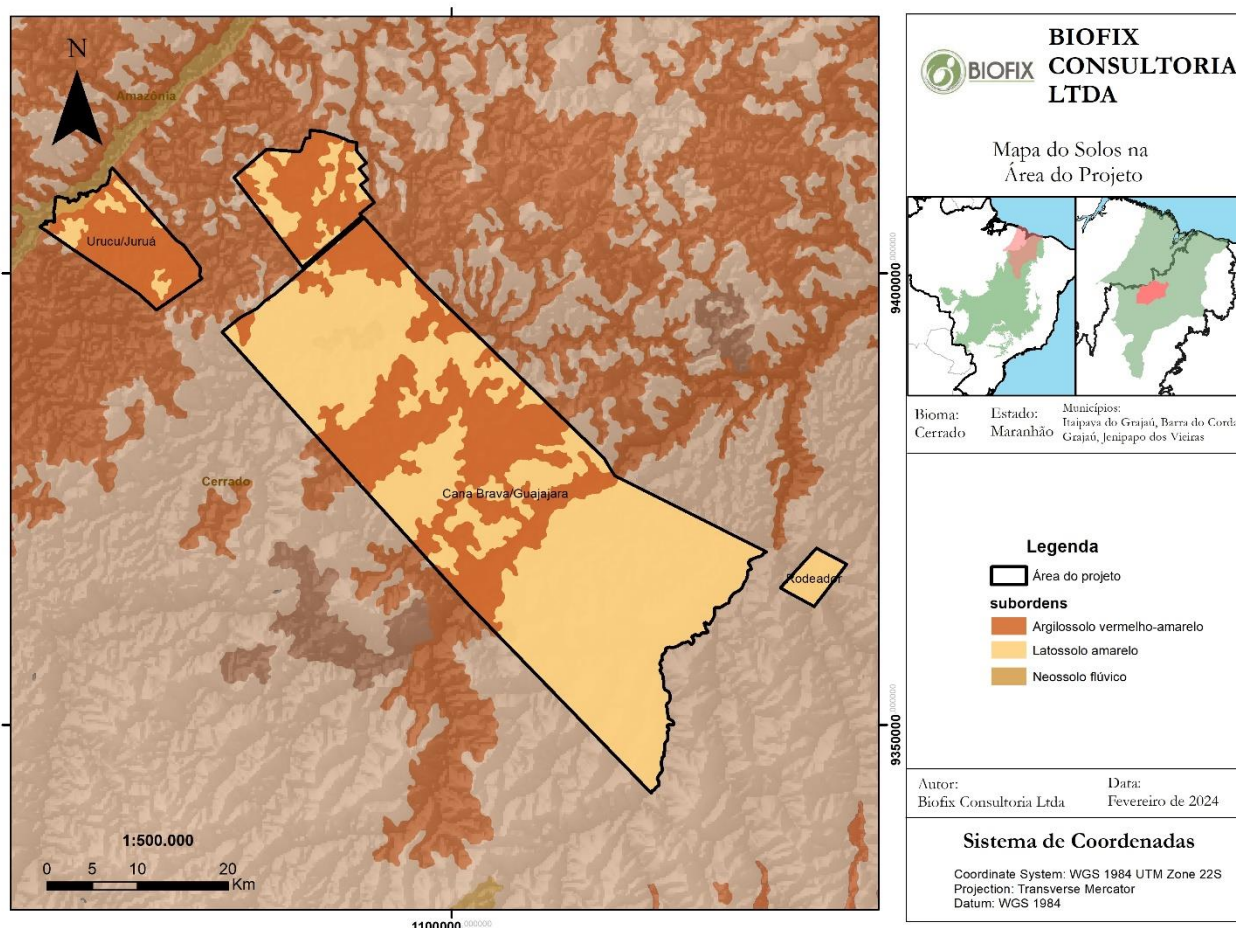
1.14.2.3 Geomorphology, edaphology and geology

Soils

According to the 2001 Brazilian Soil Map (EMBRAPA, 2001), 62.76% of the project area is located in Latosol order, yellow- latosol sub-order, which are soils made up of mineral material, clayey or very clayey, soils with more yellow color and with base saturation < 50% in most of the first 100 cm of the B horizon.

In second place, 35.7% of the project area (59,000 has) is located in Argisol orden soil, red-yellow argisol sub-order, which are soils made up of mineral material, with high and very high activity clay and combined with base saturation ≥ 50% in most of the first 100 cm of the B horizon, and with an alumina character. Finally, 769 has of the project area are located in neosols order, fluvic neosols sub-order, wich are poorly evolved soils, derived from alluvial sediments, made up of organic material less than 10 cm thick, with no type of diagnostic B horizon, and soils with high clay activity and base saturation ≥ 50% (EMBRAPA, 2018).

Figure 17. Map of soils in the project area



Source: (EMBRAPA, 2001), adapted.

Table 16. Soil order in the project area

Soil Orden	Sub-Order	Big group	Area (ha)	Area (%)
Argissolo	Red-yellow Argisols	Eutrophic Red-yellow Argisols	59,004.96	35.72
Latossolo	Yellow Latosol	Dystrophic yellow latosol	103,689.52	63.81
Neossolo	Fluvic Neosols	Eutrophic Fluvic Neosols	769.68	0.47
Total			165,205.40	100.00

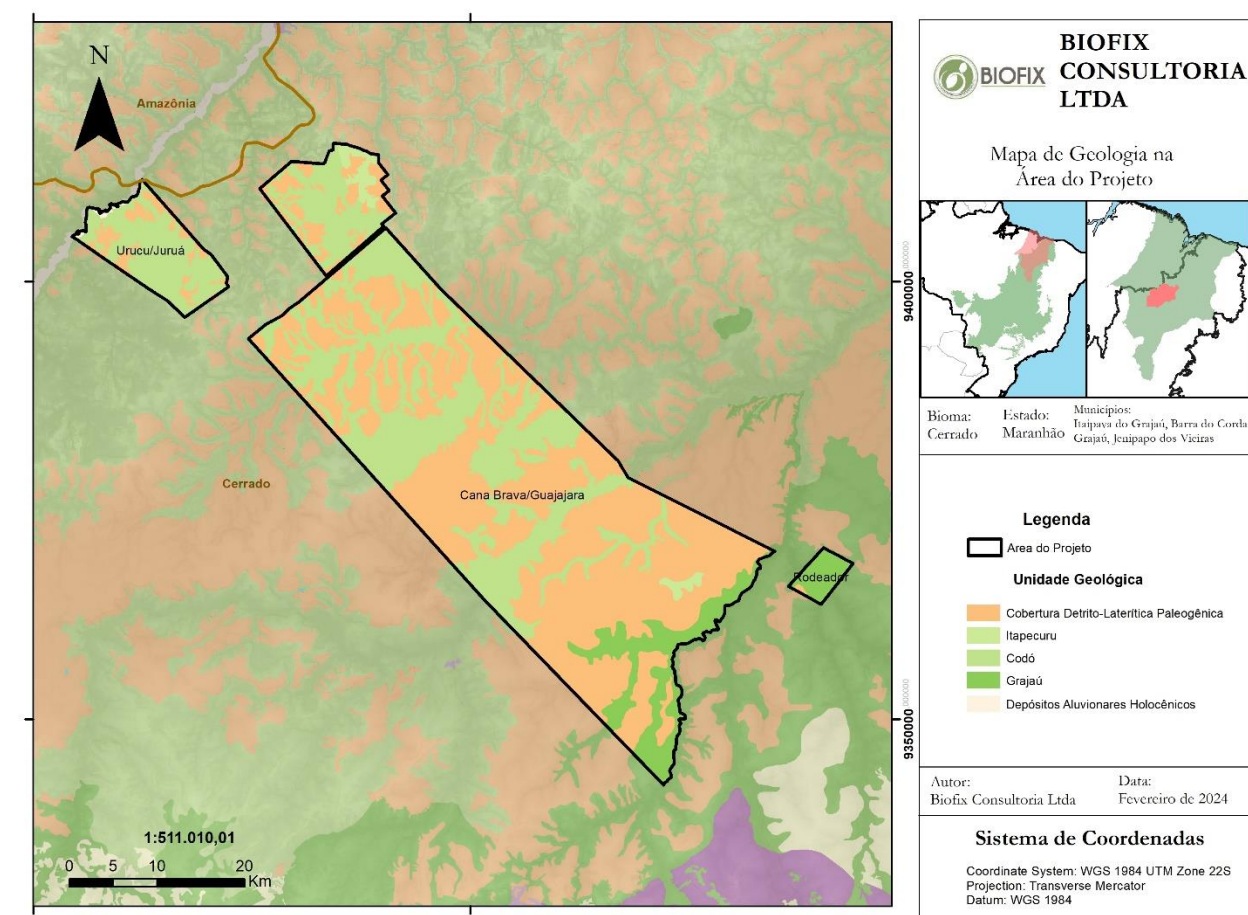
Source: (EMBRAPA, 2001), adapted.

Geology

In terms of geology, according to IBGE Natural Resources Mapping (IBGE, 2018), 5 geological units are present in the project area. The Palaeogene Detrital-Laterite Cover unit and the Holocenic Alluvial Deposits unit, which belong to geological time geological Cover and occupy 53.68% of the project area. These geological units are characterized by kaolinitic clay zone, bauxitic zone with concretions and glossitis lenses, ferruginous concretionary zone, nodular pisolitic zone, and clay capping on top (IBGE, 2011).

The geological units The Codó, Grajaú, and Itapecuru units occupy the remaining 46.33%, and which belong to the geological time Parnaíba. The formation Codó is characterized by cinza-escuras folhelhos, with intercalacoes of arenites, calcareous and glosite. On the other hand, the Grajaú formation is represented by yellowish firavels sandstones. Finally, the Itapecuru formation is made up of asphalted sandstones.

Figure 18. Map of geology in the project area



Source: (IBGE, 2018), adapted.

Table 17. Geology of the project area

Name of Geological Unit	Geological time	Name of structural province	Area (ha)	Area (%)
Palaeogene Detrital-Laterite Cover	Phanerozoic Cenozoic Paleogene Oligocene	Cenozoic cover	88,344.87	53.48
Codó	Phanerozoic Mesozoic Lower Cretaceous Aptian	Parnaíba	61,763.63	37.39
Grajaú	Phanerozoic Mesozoic Lower Cretaceous	Parnaíba	12,586.04	7.62
Itapecuru	Phanerozoic Mesozoic Lower Cretaceous Albian	Parnaíba	2,184.33	1.32
Holocenic Alluvial Deposits	Phanerozoic Cenozoic Quaternary Holocene	Cenozoic cover	324.38	0.20
Total			165,203.26	100.00

Source: (IBGE, 2018), adapted.

Geomorphology

According to IBGE Natural Resources Mapping (IBGE, 2018), the geomorphology of the project area is characterized by 4 types of geomorphological units. The Chapada de Barra do Corda unit, occupies the 81.5% of the project area, which is characterized by a tray relief (tabuleiro), with a pediplane and convex top shape.

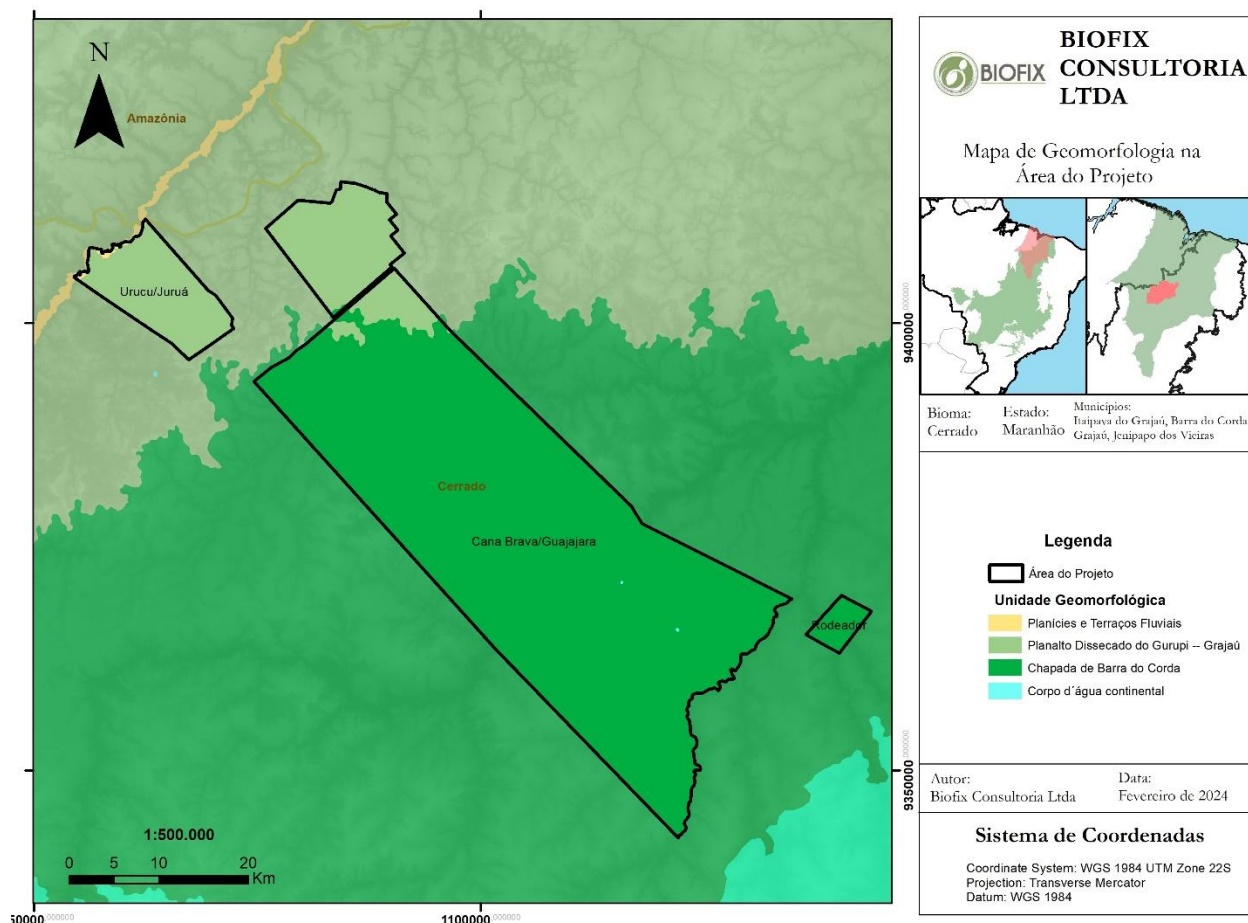
The Planalto Dissecado do Gurupi – Grajaú unit occupies 18.2% of the project area, which is characterized by patamares relief, with also a pediplane and convex top shape. The Chapada de Barra do Corda unit and the Planalto Dissecado do Gurupi – Grajaú unit, belong to the Phanerozoic Basins and Sedimentary Covers Morphostructural Domain.

Table 18. Geomorphology of the project area

Name of Geomorphical Unit	Relief compartment	Name of Morphostructural Domain	Area (ha)	Area (%)
Chapada de Barra do Corda	Tabuleiros	Phanerozoic Basins and Sedimentary Covers	134,735.77	81.56
Planalto Dissecado do Gurupi – Grajaú	Patamares	Phanerozoic Basins and Sedimentary Covers	30,115.98	18.23
Planícies e Terraços Fluviais	Planícies (plains)	Quaternary sedimentary deposits	328.90	0.20
Corpo d'água continental	Continental water body	Continental water body	21,99	0.13
Total			165,203.26	100.00

Source: (IBGE, 2018), adapted.

Figure 19. Map of Geomorphology in the project area



Source: (IBGE, 2018), adapted.

Ecosystems and vegetation

The Cerrado biome is the second largest biome in Brazil, occupying approximately 23.3% (1,983,017 km²) of the national territory and extending to the countries of Bolivia and Paraguay. In Brazil, the Cerrado covers 12 states of the country (IBGE, 2019).

Considered one of the world's biodiversity hotspots, the Cerrado has one of the highest biodiversities in the world with a high abundance of endemic species, harboring approximately 12,829 species of native plants and over 3,167 species of fauna (BFG The Brazil Flora Group, 2021).

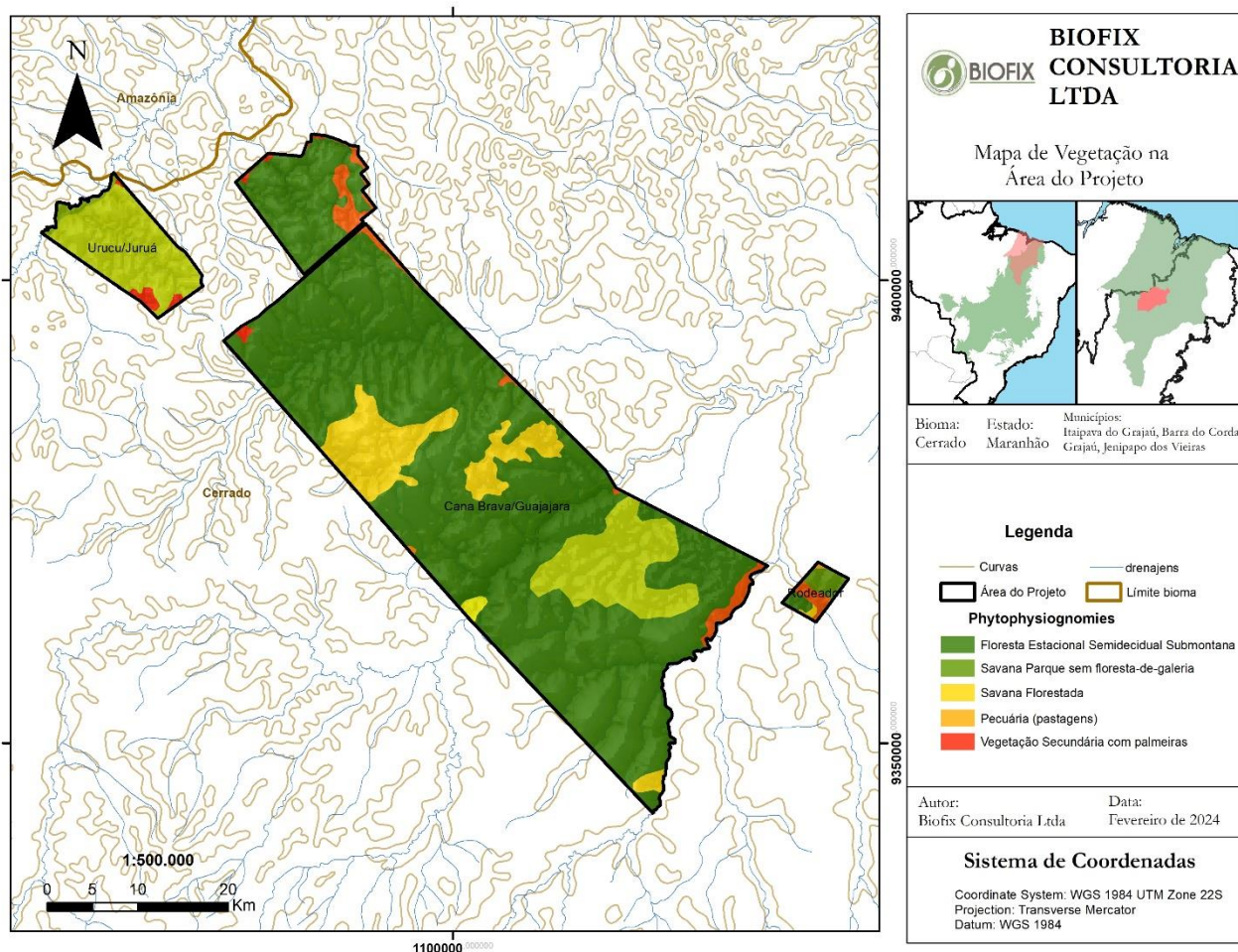
The Cerrado is a mosaic of varied vegetation and gradients of coverage and height, ranging from forests to grasslands, which is why the biome is also the second largest in terms of number of phytophysiognomies, with a total of 28 (Eiten, 1972).

According to (RIBEIRO & WALTER, 2008), the main phytophysiognomies of the biome can be grouped into three main formations: **forest formations** (riparian forest, gallery forest, dry forest, and cerradão), **savanna formations** (dense cerrado, typical cerrado, sparse cerrado, rocky cerrado, vereda, cerrado park,

and palm grove), and **grassland formations** (dirty field, clean field, and rocky field). Approximately 35% of the biome is represented by the park savanna and woody-grass savanna phytophysionomies, savanna formations with predominantly herbaceous vegetation.

The KA'A PYAHU REDD+ project is located in the state of Maranhão, in the northeast region of Brazil, in a transition zone between the Cerrado and Amazon biomes (MMA, IBAMA, 2000), which is also classified as an ecotone, or zone of ecological tension between adjacent communities or biomes that undergo mutual interference, where there is a high genetic diversity (HAIDAR, et al., 2013).

Figure 20. Vegetation and phytophysionomy map in the project area.



Source: (IBGE, 2018), adapted.

According to the vegetation map of Brazil (IBGE, 2018), 87.63% of the vegetation area is composed of Submontane semi Deciduous Seasonal Forest and 8.69% is savanna.

Abv.	Phytophysiognomy	Area (ha)	Area (%)
Fs	Submontane Semi Deciduous Seasonal Forest	144771,59	87.63
Sd	Forested Savanna	14368,01	8.69
Ap	Grasslands	4567,22	2.76
Vsp	Secondary vegetation	1501,60	0.91
	Total	165,208.42	100.00

Coverage and land use

BIOFIX CONSULTORIA LTDA

Mapa de cobertura e uso da terra na Área do Projeto

Legenda

- Limite bioma
- curva_Maria
- drenagem_Maria
- Área do projeto

1. Forest

- 1.1-Forest Formation-3
- 1.2 Savanna Formation-4

2. Non Forest Natural Formation

- 2.2. Grassland-12

3. Farming

- 3.1. Pasture-15
- 3.2.1.1. Soybean-39
- 3.2.1.5. Other Temporary Crops-41
- 3.4. Mosaic of Uses-21

4. Non vegetated area

- 4.2. Urban Area-24

5. Water

- 5.1. River, Lake and Ocean-33

Sistema de Coordenadas

Coordinate System: WGS 1984 UTM Zone 22S
Projection: Transverse Mercator
Datum: WGS 1984

⁷ <https://brasil.mapbiomas.org/colecoes-mapbiomas/>

In terms of land cover, according with the data of Mapbiomas, approximately 90% of the project area is covered by forest formations. The remaining percentage is covered by savanna formations, pastures and grasslands.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

Legal and Documentary Aspects of the KAA PYAHU REDD+ Project

The KAA PYAHU REDD+ Project is developed in compliance with local and national laws, statutes and regulations relating to environmental projects, REDD+ and climate change. Importantly, the formulation of the project is aligned with compliance with national and international legislation. This approach ensures strict adherence to regulations, promoting the effectiveness and legitimacy of the project in the legal and regulatory context.

Legislation:

International

UN Declaration on the Rights of Indigenous Peoples and ILO Convention 169

Federal Constitution

Articles 231 and 232 of the Federal Constitution provide for the recognition of the social organization, customs, languages, beliefs and traditions of indigenous communities, as well as the right to the lands they traditionally occupy and the usufruct of their wealth.

Federal legislation

The National Policy for Territorial and Environmental Management of Indigenous Lands - PNGATI, established by Decree 7.747/2012, has the following objectives: "to promote the participation of indigenous peoples, communities and organizations in environmental and territorial protection actions on indigenous lands", as well as "to promote actions aimed at recovering and restoring degraded areas on indigenous lands".

LAW No. 6.001/73 - Indian Statute

LAW No. 5.371/67 - establishing FUNAI

State of Maranhão

LAW No. 9.279 OF OCTOBER 20, 2010 (Establishes the State Environmental Education Policy and the State Environmental Education System of Maranhão)

LAW No. 9.370, OF MAY 13, 2011 (Adds item IV to art. 1 of Law No. 4.734/86, which prohibits the felling of babassu and makes other provisions)

LAW No. 9413, OF JULY 13, 2011 (Regulates art. 241 of the Constitution of the State of Maranhão, Chapter III, Section VII of State Law No. 5405, of April 8, 1992, Chapter II, Section VIII of State Decree No. 13494, of November 12, 1993, and establishes the State System of Nature Conservation Units of Maranhão and makes other provisions)

LAW No. 10.169 OF DECEMBER 5, 2014 (Provides for the protection of all animals at the state level)

LAW No. 10.595, OF MAY 24, 2017 (Establishes the Green Maranhão Program, aimed at fostering and developing projects to support environmental conservation and recovery)

LAW No. 11.365, OF OCTOBER 19, 2020 (Creates and organizes the Environmental School of the State of Maranhão, and makes other provisions.

LAW No. 11.468, OF MAY 7, 2021 (Establishes the Green Life Seal, to be awarded to private companies that prove they have adopted environmental sustainability practices in the state of Maranhão, and makes other provisions)

LAW Nº 11578 OF 01/11/2021 - Establishes the Policy for Reducing Greenhouse Gas Emissions from Deforestation and Forest Degradation, the Conservation of Forest Carbon Stocks, Sustainable Forest Management and Increased Forest Carbon Stocks (REDD+), Environmental Asset Management and Payment for Environmental Services (PES) in the State of Maranhão, called the Jurisdictional REDD+ and PES System, and amends State Law No. 11. 000, of April 2, 2019, to expand the scope of action of Maranhão Parcerias - MAPA.

"IX - Private Projects Incentive and Harmonization Program;"

DECREE No. 37969 OF 10/26/2022 - Regulates State Law No. 11. 578, of November 1, 2021, which instituted the Policy for Reducing Greenhouse Gas Emissions from Deforestation and Forest Degradation, Conservation of Forest Carbon Stocks, Sustainable Forest Management and Increasing Forest Carbon Stocks (REDD+), Management of Environmental Assets and Payment for Environmental Services (PES) in the State of Maranhão, called the Jurisdictional REDD+ and PES System.

"IX - Private Project Incentive and Harmonization Program: aims to encourage and promote private sector investment in projects that could potentially integrate, align and harmonize with the state REDD+ system, facilitating the formulation and implementation of REDD+ promotion, financing, investment and methodology rules by the private sector, contributing to the objectives of this Decree, with the conditions and rules to be applied respecting national legislation, Brazil's international commitments and state rules;"

Legal Requirements (Compliance with Legal Aspects)

The company BIOFIX Consultoria Ltda., as the structuring entity, will conduct an annual assessment of compliance with legal requirements, addressing:

a. Identification of legal and environmental requirements.

- b. Adjustment of the reference level in accordance with the latest updates to the Forestry Emissions Reference Level (FREL).
- c. Verification of compliance with the new Land Use Schemes in the municipalities of Jenipapo dos Vieiras, Barra do Corda, Grajaú and Itaipava do Grajaú.
- d. Identifying social and environmental safeguards and ensuring their compliance.

Safeguards

Articulation with the Cancún Social and Environmental Safeguards

Context of Compliance with Social and Environmental Safeguards

Social and environmental safeguards originate from the mitigation actions of the United Nations Framework Convention on Climate Change (UNFCCC), defining measures to reduce emissions associated with deforestation and forest degradation, conservation and increasing forest carbon stocks (REDD+). The Cancun Safeguards were established to guide countries implementing REDD+.

The UNFCCC REDD+ Safeguards outline a global framework of social, environmental and governance principles. By complying with this global framework, countries can minimize risks and increase benefits, both carbon and non-carbon. They apply to all REDD+ initiatives, adapted to local, regional and national scales.

The UNFCCC establishes that the implementation of measures to reduce emissions from deforestation and forest degradation must follow the "Cancun Safeguards". These safeguards include complementary actions with national forest program objectives, transparency in forest governance structures, respect for the rights of indigenous peoples, effective stakeholder participation, and consistency with the conservation of natural forests and biological diversity, actions to avoid risks of reversals of REDD+ results, and actions to reduce the displacement of carbon emissions.

Management of Legal Requirements (Safeguards A, B) - Property Rights and Land Tenure

The 4 territories have an area demarcated by Decree, as follows:

- Cana Brava - Homologated by Decree 246 - 30/10/1991.
- Lagoa Comprida - Homologated by Decree 313 - 30/10/1991.
- Urucu-Juruá - Approved by Decree 382 - 26/12/1991.
- Rodeador - Homologated by Decree 88.813 - 06/10/1983

Socialization and CPLI (Safeguards B, C, D) - Legal and Jurisprudential Framework

The KAA PYAHU project went through a process of socialization and prior, free and informed consultation (CPLI) with the indigenous communities, ensuring that their rights were respected. The consultation was carried out in accordance with international standards and Brazilian legislation. The communities' acceptance was formalized in General Assemblies.

The KAA PYAHU REDD+ Project complies with the Cancun Safeguards established by the UNFCCC. The effective management of legal requirements and the implementation of socio-environmental safeguards guarantee the mitigation of adverse impacts, promoting environmental, social and economic benefits for the communities involved and contributing positively to the fight against climate change.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Is the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes

☒ No

1.18 Sustainable Development Contributions

1.18.1 Sustainable Development Contributions Activity Description

The lines of action were designed to improve the general quality of life of the community belonging to the indigenous territories. With the implementation of the REDD+ project, it is hoped that there will be significant contributions to sustainable development, according to each territory.

Institutional and governance strengthening: REDD+ training workshops for leaders and communities, territorial monitoring - Internal Management Plan for the territories, equipment to strengthen the associations, construction of cultural houses, support for participation in regional and national indigenous mobilizations.

Strengthening Education: Building and improving the school structure in the communities, strengthening the mother tongue. Strengthening local culture in schools (handicrafts, traditional crafts, indigenous school education - indigenous education), workshops on traditional cultural activities with teachers from the community), support for indigenous pupils in the municipality to continue their school education, strengthening of traditional festivals.

Participatory reforestation of degraded ecosystems: Irrigation structure, distribution of water for the growth of plantations, planting of native species and training of local agents and forest guardians.

Monitoring and control of deforestation and forest degradation: Territorial surveillance, prevention of fires during critical periods, monitoring of harassment and threats of theft of internal natural resources.

Green businesses: Beekeeping, native seed handicrafts, fish farming, native seedling nurseries, native fruit growing.

Infrastructure: Treatment of drinking water, creation of cooperatives to strengthen internal production activities, construction of a health structure, construction of a flour mill, construction of housing structures and construction of roads.

Strengthening Health: Strengthening traditional medicine, collecting garbage and waste that remains in the territory and generates disease - prevention, planting native species that are used in traditional medicine.

The expectation is that the implementation of the REDD+ project in the Guajajara communities will bring improvements in livelihoods, education, health and environmental preservation efforts, ultimately contributing to sustainable development. The table below describes how the project's activities hope to contribute to the Sustainable Development Goals.

Table 20. National targets related to Sustainable development goals and REDD+ project

SDG	Brazilian targets
6	6.3 By 2030, improve water quality by reducing pollution, eliminating dumping and minimizing the release of chemicals and hazardous materials, halving the proportion of untreated wastewater and substantially increasing recycling and safe reuse globally.
	6.b Support and strengthen the participation of local communities to improve water and sanitation management.
10	10.2 By 2030, empower and promote the social, economic and political inclusion of all, regardless of age, gender, disability, race, ethnicity, origin, religion, economic or other condition.
13	13.b Promote mechanisms for building capacity for climate change-related planning and effective management in least developed countries, including with a focus on women, youth, local and marginalized communities.
14	14.4 By 2020, effectively regulate harvesting, and end illegal, unreported and unregulated overfishing and destructive fishing practices, and implement science-based management plans, to restore fish populations in the shortest possible time, at least at levels that can produce maximum sustainable yield, as determined by its biological characteristics.
15	15.a Mobilize and significantly increase, from all sources, financial resources for the conservation and sustainable use of biodiversity and ecosystems.

Source: (ONU BR – NAÇÕES UNIDAS NO BRASIL – ONU BR., 2015).

1.18.2 Sustainable Development Contributions Activity Monitoring

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
<i>Sequential row number</i>	<i>SDG Target number</i>	<i>Number and text of SDG indicator or, if no official SDG indicator is applicable, user-defined indicator</i>	<i>Indicate the project's contribution to the SDG Indicator (implemented activities to increase or decrease)</i>	<i>Brief description of the quantifiable impact of the project's activities related to the SDG indicator, during the monitoring period.</i>	<i>Brief description of the cumulative quantifiable impact of the project's activities related to the SDG indicator, over the project lifetime.</i>
1)	1.1	1.1.1 Proportion of population below the international poverty line	Implemented activities to decrease	No further changes this monitoring period	The project has increased the 65 participants' total daily income from 1.20 USD/day to 2.57 USD/day, bringing them above the international poverty line
2)	3.2	3.3.3 Malaria incidence per 1,000 population	Implemented activities to decrease	Lowered the malaria incidence per 1,000 to 98 by distributing 200 additional bed nets and conducted malaria prevention workshops.	Lowered the malaria incidence per 1,000 from 157 to 98
3)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	By conserving 400 ha of tropical rainforest, Project X has prevented the release of 250 thousand tonnes of carbon into the atmosphere during the monitoring period	Prevented the release of 750 thousand tonnes of carbon into the atmosphere

4)	6.1	Proportion of the rural population who have easy access to a safe water supply	Implemented activities to increase	Completed construction of 4 additional improved wells to provide potable water to 230 people	Provided at least 10 liters of potable water per day to 1,200 people, a 40% increase in the catchment area, over the project lifetime by constructing improved wells

1.19 Additional Information Relevant to the Project

Leakage Management

Where applicable, describe the leakage management plan and implementation of leakage and risk mitigation measures.

Commercially Sensitive Information

Indicate whether any commercially sensitive information has been excluded from the public version of the project description using Appendix 1, and briefly describe the items to which such information pertains. Provide justification for why the information is commercially sensitive and confirm that it is not otherwise publicly available.

Note - Information related to the determination of the baseline scenario, demonstration of additionality, and estimation and monitoring of GHG emission reductions and removals (including operational and capital expenditures) cannot be considered to be commercially sensitive and must be provided in the public versions of the project documents.

Further Information

Include any additional relevant legislative, technical, economic, sectoral, social, environmental, geographic, site-specific and/or temporal information that may have a bearing on the eligibility of the project, the GHG emission reductions or carbon dioxide removals, or the quantification of the project's reductions or removals.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Use the table below to describe the stakeholder identification process. Where the rows do not apply, provide justification in the cell in the table below.

Stakeholder Identification	Describe the process(es) used to identify stakeholders likely impacted by the project. List the stakeholders identified.
----------------------------	--

Legal or customary tenure/access rights	<i>Describe any legal or customary tenure/access rights to territories and resources, including collective and conflicting rights, held by stakeholders, indigenous people (IPs), local communities (LCs), and customary rights holders.</i>
Stakeholder diversity and changes over time	<i>Describe the social, economic and cultural diversity within stakeholder groups, the differences and interactions between the stakeholder groups, and any changes in the make-up of each group over time.</i>
Expected changes in well-being	<i>Describe the expected changes in well-being and other stakeholder characteristics relative to the baseline scenario, including changes to ecosystem services identified as important to stakeholders;</i>
Location of stakeholders	<i>Describe the location of stakeholders, IPs, LCs, and customary right holders, and areas outside the project area that are predicted to be impacted by the project.</i>
Location of resources	<i>Describe the location of territories and resources which stakeholders own or to which they have customary access.</i>

2.1.2 Stakeholder Consultation and Ongoing Communication

Use the table below to describe the process for and the outcomes from the stakeholder consultation conducted prior to project initiation.

Date of stakeholder consultation	<i>DD-Month-YYYY</i>
Stakeholder engagement process	<i>Describe the process to engage stakeholders in a culturally appropriate manner (e.g., dates of announcements or meetings, language and gender sensitivity). Describe the process or methods used to document the outcomes.</i>
Consultation outcome	<i>Summarize the discussion around consent to project design and implementation, risks, costs and benefits of the project, all relevant laws and regulations covering workers' rights in the host country, the discussion of FPIC, and the VCS validation and verification process.</i>

Ongoing communication	Describe the mechanisms for ongoing communication with stakeholders.
Stakeholder input	Describe how due account was taken of all input received during the consultation. Include details on any updates to the project design or justify why updates were not necessary or appropriate.

2.1.3 Free Prior and Informed Consent

Use the table below to describe the outcome of the FPIC process as part of the stakeholder consultation process at validation and during the monitoring period.

Obtaining consent	Describe and demonstrate how consent to implement the project activities was obtained from those concerned, including IPs, LCs, and customary rights holders, and a transparent agreement was reached. Describe any ongoing or unresolved conflicts and demonstrate that the project does not exacerbate nor influence the outcomes of unresolved conflicts.
Outcome of FPIC	Describe the outcome of the FPIC process, the transparent agreement, and the information disclosed prior to establishing a transparent agreement with those concerned, IPs, LCs, and customary rights holders. Provide assurance that the project has not encroached on land, relocated people without consent, and forced physical or economic displacement

2.1.4 Grievance Redress Procedure

Use the table below to describe the grievance redress procedures developed to resolve any conflicts which may arise between the project proponent and stakeholders.

Development process	Describe the process used to develop the grievance redress procedure including processes for receiving, hearing, responding, and attempting to resolve grievances within a reasonable time period, taking into account culturally appropriate conflict resolution methods.
Grievance redress procedure	Describe the grievance redress procedures developed with stakeholders.

Use the table below to describe any grievances that were raised during the monitoring period and the steps the project proponent took to resolve the grievance including the resolution of the grievance. Repeat the rows as necessary. Where no grievances were raised, indicate this with NA. and demonstrate that the procedure is easily accessible to stakeholders for ongoing consultation.

Grievances received	Resolution and outcome
Summarize the grievance raised during the monitoring period.	Describe the steps taken to resolve the grievance including the outcomes of the resolution.
	...

2.1.5 Public Comments

Summarize any public comments submitted during the public comment period and any comments received after the public comment period. Demonstrate how due account was taken of all comments received. Include details on when the comments were received, and any updates to the project design or demonstrate the insignificance or irrelevance of comments.

Comments received	Actions taken
Summary of comment received	Provide a summary of actions taken and any project design updates, or justify why updates were not necessary or appropriate.
...	

2.2 Risks to Stakeholders and the Environment

Use the table below describe the risk assessment and outcome of the potential risks to stakeholders and the environment. Describe the mitigation or preventative measure in place to prevent or mitigate the risk. Where no risk is identified, write "No risk identified" in the first column, and provide justification in the second column.

	Risks identified	Mitigation or preventative measure taken
Risks to stakeholder participation		
Working conditions		

Safety of women and girls		
Safety of minority and marginalized groups, including children		
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)		

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

Use the table below to demonstrate the project's respect for rights related to work and labor.

Discrimination and sexual harassment	Demonstrate that no discrimination or sexual harassment has occurred or will occur.
Management experience	Demonstrate that management teams have expertise or experience in implementing similar project activities and engaging communities. Where relevant experience is lacking, demonstrate how the project proponent has partnered with other organizations to support the project or have a recruitment strategy to fill the identified gaps.
Gender equity in labor and work	Demonstrate that equal opportunities have been or will be provided in the context of gender equity and pay for labor and work.
Human trafficking, forced labor, and child labor	Demonstrate that the project does not and will not use victims of human trafficking, forced labor, and child labor.

2.3.2 Human Rights

Demonstrate how the project recognizes, respects, and promotes the protection of the rights of IPs, LCs, and customary rights holders in line with applicable international human rights law, and the United Nations Declaration on the Rights of Indigenous Peoples and ILO Convention 169 on Indigenous and Tribal Peoples.

2.3.3 Indigenous Peoples and Cultural Heritage

Demonstrate that the project preserves and protects cultural heritage as part of project activities.

2.3.4 Property Rights

Using the table below, describe the property rights of IPs, LCs, and customary rights holders and demonstrate respect of such rights.

Rights to territories and resources	<i>Describe any legal or customary tenure/access rights to territories, property, and resources, including collective and/or conflicting rights, held by stakeholders.</i>
Respect for property rights	<i>Describe the measures implemented to protect and preserve the property rights of IPs, LCs, and customary rights holders.</i>

2.3.5 Benefit Sharing

Where the project impacts property rights as described in Section 2.4.4 above, use the table below to describe the project's benefit sharing agreement.

Process used to design the benefit sharing plan	<i>Describe the process used to develop the benefit-sharing agreement with the affected stakeholder groups.</i>
Summary of the benefit sharing plan	<i>Describe the benefit-sharing agreement including provisions for renegotiation. Where affected stakeholder groups wish to keep elements of the plan private, provide the full arrangement as a commercially sensitive document. The project proponent shall demonstrate that the community wishes to keep this information private.</i>
Approval and dissemination of benefit sharing plan	<i>Demonstrate that the benefit-sharing agreement was agreed up on by the affected stakeholder groups, and that the agreement was shared in a culturally appropriate manner. Demonstrate that the agreement is readily accessible should stakeholders wish to review the agreement.</i>
Benefit sharing during the monitoring period	<i>Describe the implementation of the benefit sharing plan.</i>

2.4 Ecosystem Health

Identify and summarize any risks to the environment and the steps taken to mitigate them. Where no risk is identified, write "No risk identified" in the first column, and provide justification in the second column.

	Risks identified	Mitigation or preventative measure taken
Impacts on biodiversity and ecosystems		
Soil degradation and soil erosion		
Water consumption and stress		
Usage of fertilizers		

2.4.1 Rare, Threatened, and Endangered species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☐ No

If yes, list such species and habitats in the table below and provide evidence that the project will not adversely impact these areas.

Species and habitat	Demonstrate that the project will not adversely impact habitats for rare, threatened, or endangered species.
...	...

2.4.2 Introduction of species

Demonstrate, using the table below, that no invasive species will be used as part of project activities. Categorize each species as native, non-native, and indicate if the species is a mono-culture. Where the species is non-native include an explanation of possible adverse effects of its usage, and a description of how the project will mitigate such risks and has mitigated such risks during the monitoring period. This table is not required for projects with no planting or species introduction; this section may be indicated as N/A.

Species introduced	Classification	Justification for use	Adverse effects and mitigation

Where invasive species exist in the project area, list such species in the table below and demonstrate that the project activity will not allow the species to thrive.

Existing invasive species	Mitigation measures to prevent spread or continued existence of invasive species

2.4.3 Ecosystem conversion

ARR, ALM, WRC or ACoGS projects shall provide evidence that the project area was not cleared or drained of existing natural ecosystems, unless such clearing took place at least 10 years prior, or the dominant land cover was invasive.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Table 21. Methodologies, Modules and tools for the KA'A PYAHU REDD+ project

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VM0007	REDD+ Methodology Framework (REDD+MF).	7.0
Module	VMD0001	Estimation of carbon stocks in the above- and belowground biomass in live tree and non-tree pools (CP-AB)	1.2
Module	VMD0007	Estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation and unplanned wetland degradation (BL-UP)	3.3
Module	VMD0010	Estimation of emissions from activity shifting for avoiding unplanned deforestation and avoiding unplanned wetland degradation	1.2
Module	VMD0015	Methods for monitoring of GHG emissions and removals in REDD and CIW projects (M-REDD)	2.2

Module	VMD0016	Methods for stratification of the project area (X-STR)	1.3
Module	VMD0017	Estimation of uncertainty for REDD project activities (X-UNC)	2.2
Tool	VT0001	Tool for the demonstration and Assessment of additionality in vcs Agriculture, Forestry and Other land Use (AFOLU) project activities	3.0

3.2 Applicability of Methodology

Table 22. Specific applicability of methodologies, modules, and tools in REDD activities

Methodology ID	Applicability condition	Justification of compliance
VM0007	-Land in the project area has qualified as forest (following the definition used by VCS; in addition, see Section 5.1.2) for at least the 10 years prior to the project start date. Mangrove forests are excluded from any tree height requirement in a forest definition, as they consist of (close to) 100% mangrove species, which often do not reach the same height as other tree species and occupy contiguous areas and their functioning as a forest is independent of tree height. -Baseline deforestation and forest degradation in the project area fall within one or more of the following categories: Unplanned deforestation (VCS category AUDD)	The analysis of land use categories in the project is based mainly on Mapbiomas which utilizes a satellite imagery dataset used composed by Landsat 5 (Thematic Mapper), Landsat 7 (Enhanced Thematic Mapper Plus) and Landsat 8 (Operational Land Imager), Landsat sensors. The MapBiomas classification scheme is a hierarchical system with a combination of LULC classes compatible with the Food and Agriculture Organization (FAO) and IBGE classification systems. -The procedures finding into the boundaries of the project are in the category of Unplanned deforestation (AUDD).

	- Planned deforestation/degradation (VCS category APD) -Baseline agents of deforestation must: - clear the land for tree harvesting, settlements, crop production (agriculturalist) or ranching or aquaculture, where such clearing for crop production or ranching or aquaculture does not amount to large scale industrial agriculture or aquaculture activities - have no documented and uncontested legal right to deforest the and for these purposes; and - be either residents in the reference region for deforestation or immigrants.	- The agents of deforestation that threaten the indigenous territories and the project proponents, are the result of human actions Illegal logging: One of the agents of deforestation is the illegal exploitation of timber resources. The invasion of territories in search of wood for illegal trade, threatens natural resources and also the lives of people in the territory. Through episodes of conflict, threats and even death, making it one of the reports of violence against indigenous peoples in Brazil produced by the Indigenous Missionary Council, an important document and instrument of national denunciation (Annex 019) The Guajajara people report continued invasions by loggers and farmers who produce soy, causing socio-environmental damage. There are also reports of wild animal hunters, who threaten communities. The illegal removal of wood from the Urucu Juruá Indigenous Land has been taking place since September 2019 and according to estimates by the Guajajara, around 70% of the forest in the Indigenous Land has already been destroyed. Illegal Pastures: There are reports of illegal land leasing and illegal pastures in the territories of the project proponents. Around 75% of
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		the forest cover in the state of Maranhão was deforested to provide space for agriculture and livestock, points out the National Institute for Space Research, INPE. Approximately 70% of the remaining native forest is found in the 16 indigenous lands and conservation units of Maranhão. What makes areas attractive for extractivism.
VMD0001	This module is applicable to all forest types and age classes.	For the analysis of the project, the Mapbiomas⁸ data are used. In the case of forest cover, this classification covers the different forest phytophysionomies for each biome. These phytophysionomies correspond to the ecosystems presented in the NREF⁹ Brazil, and their equivalence with FAO.
VMD0007	The module must be applied to all project activities where the baseline agents of deforestation: (i) clear the land for tree harvesting, settlements, crop production (agriculturalist) or ranching or aquaculture, where such clearing for crop production or ranching or aquaculture does not amount to large scale industrial agriculture or aquaculture activities. (ii) have no documented and uncontested legal right to deforest the and for these purposes; (iii) and be either residents in the reference region for deforestation or immigrants.	The project complies with the applicability conditions expressed for the VM0007 methodology.

⁸ <https://brasil.mapbiomas.org/codigos-de-legenda/>

⁹ <https://redd.unfccc.int/submissions.html?country=bra>

VMD0010	Applicable to estimate changes in carbon stocks and emissions due to displacement of activities that cause deforestation of land. Activities subject to displacement are the conversion of forest land to other land uses. Mandatory module if BL-UP was used.	The module VMD007 (BL-UP) is mandatory for the project activity (unplanned deforestation). Therefore, it is mandatory to apply module VMD0010 to determine emissions in the leakage belt. In addition, the conditions for feasible scenarios of displacement of emissions to the leakage belt are fulfilled.
VMD0015	Strata as defined in the relevant baseline modules are fixed and may not be changed without baseline revision. The module is mandatory for REDD, CIW-REDD, RWE-REDD and stand-alone CIW project activities.	This module is mandatory because it is a REDD+ project.
VMD0016	The project area may be stratified ex ante, and this stratification may be revised ex post for monitoring purposes. Established strata may be merged if reasons for their establishment have disappeared or have proven irrelevant to key variables for estimating net GHG emissions or removals. A map displaying the final delineation of strata must be included in the project description. Areas of individual strata naturally sum to the total project area; any discrepancies must be reconciled.	The forest strata are defined according to the NREF Brazil forest types. In the case of the project, the forest strata correspond to the transitional forest phytophysiognomies between the Cerrado and Amazon biomes.
VMD0017	This module is mandatory when using methodology REDD+ MF. It is applicable for estimating the uncertainty of estimates of emissions and removals of CO₂-e	The analysis of the project data uses Mapbiomas as a source, which works with a confidence interval of 95%, and a maximum standard error of 5% (Souza, et al., 2020).

generated from REDD and WRC project activities.

3.3 Project Boundary

Define the project boundary and identify the relevant GHG sources, sinks and reservoirs for the project and baseline scenarios (including leakage if applicable). Add rows as needed.

Source		Gas	Included?	Justification/Explanation
Baseline	Source 1	CO ₂		
		CH ₄		
		N ₂ O		
		Other		
	Source 2	CO ₂		
		CH ₄		
		N ₂ O		
		Other		
Project	Source 1	CO ₂		
		CH ₄		
		N ₂ O		
		Other		
	Source 2	CO ₂		
		CH ₄		
		N ₂ O		
		Other		

Provide a diagram or map of the project boundary, clearly showing the physical locations of the various installations or management activities taking place as part of the project activity based on the description provided in Section 1.12 (Description of the Project Activity) above.

For non-AFOLU projects, include in the diagram the equipment, systems and flows of mass and energy. Include the GHG emission sources identified in the project boundary.

For AFOLU projects, include in the diagram or map the locations of where the various measures are taking place, any reference areas and leakage belts.

3.4 Baseline Scenario

Identify and justify the baseline scenario, in accordance with the procedure set out in the applied methodology and any relevant tools. Where the procedure in the applied methodology involves several steps, describe how each step is applied and clearly document the outcome of each step.

Explain and justify key assumptions, rationale, and methodological choices. Provide all relevant references.

3.5 Additionality

Demonstrate and assess the additionality of the project, in accordance with the applied methodology and any relevant tools, taking into account the following additionality methods:

3.5.1 Regulatory Surplus

Is the project registered or seeking registration in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country

☐ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes

☐ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes

☐ No

If no, describe which mandated laws, statutes, or other regulatory frameworks require project activities and provide evidence of systematic non-enforcement to demonstrate regulatory surplus.

3.5.2 Additionality Methods

- *Where a project method is applied to demonstrate additionality and the procedure in the applied methodology or tool involves several steps, describe how each step is applied and clearly document the outcome of each step. Indicate clearly the method selected to demonstrate additionality (e.g., investment analysis or barrier analysis in the case of the CDM Tool for the demonstration and assessment of additionality). Where barrier analysis, or equivalent, is used to demonstrate additionality, only include the most relevant barriers. Justify the credibility of the barriers with key facts and/or assumptions and the rationale. Provide all relevant references.*
- *Where a performance method is applied to demonstrate additionality, demonstrate that performance can be achieved to a level at least equivalent to the performance benchmark metric.*
- *Where the methodology applies an activity method for the demonstration of additionality, include a statement that notes that conformance with the positive list is demonstrated in the Applicability of Methodology section above.*
- *Provide sufficient information (including all relevant data and parameters, with sources) so that a reader can reproduce the additionality analysis and obtain the same results.*

3.6 Methodology Deviations

Describe and justify any methodology deviations applied, including any previous deviations. Include evidence to demonstrate the following:

- *The deviation will not negatively impact the conservativeness of the quantification of GHG emission reductions or removals.*
- *The deviation relates only to the criteria and procedures for monitoring or measurement and does not relate to any other part of the methodology.*

4 IMPLEMENTATION STATUS

4.1 Implementation Status of the Project Activity

For all projects, describe the implementation status of the project activity(s), including information on the following:

- *The operation of the project activity(s) during the monitoring period, including any information on events that may impact the GHG emission reductions or removals and monitoring.*
- *Any other changes (e.g., to project proponent or other entities).*

For AFOLU projects, include information on the following:

- Where no new project activities have been implemented during the current monitoring period, demonstrate that previously implemented project activities continued to be implemented during the current monitoring period.
- Report any loss of carbon stock that occurred during the monitoring period. The date of the loss(es), date of discovery(s), size (hectares impacted) and extent (tCO₂e) of the loss must be described. Specify if the loss meets the definition of a loss event and/or reversal. In all cases, justify how the project meets VCS requirements related to loss events and reversals.

5 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

Describe the procedure for quantification of baseline emissions and/or carbon stock changes in accordance with the applied methodology. Baseline emissions may be negative where carbon stock increases (sinks) exceed baseline emissions. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculations. Explain and justify all relevant methodological choices (e.g., with respect to selection of emission factors and default values). Include all calculations in the emission reduction and removal calculation spreadsheet.

5.2 Project Emissions

Describe the procedure for quantification of project emissions and/or carbon stock changes in accordance with the applied methodology. Project emissions may be negative where carbon stock increases (sinks) exceed project emissions. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculations. Explain and justify all relevant methodological choices (e.g., with respect to selection of emission factors and default values). Include all calculations in the emission reduction and removal calculation spreadsheet.

5.3 Leakage Emissions

Describe the procedure for quantification of leakage emissions in accordance with the applied methodology. Specify the reductions and removals separately where the applied methodology

provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculations. Explain and justify all relevant methodological choices (e.g., with respect to selection of emission factors and default values). Include all calculations in the emission reduction and removal calculation spreadsheet.

5.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

Describe the procedure for the quantification of estimated GHG emission reductions (reductions) and carbon dioxide removals (removals). Include all relevant equations.

For data and parameters monitored, use the estimated data/parameter values provided in Section 6.2 below. Document how each equation is applied, in a manner that enables the reader to reproduce the calculations. Provide calculations for all key equations to allow the reader to reproduce the annual calculations for estimated reductions or removals. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all of the above in the emission reduction and removal calculation spreadsheet.

Complete the tables below by vintage period (calendar year). Note that the baseline or project emissions subtotals may be negative where sinks exceed emissions. Only specify the estimated VCUs for reductions and removals separately where the applied methodology provides procedures and equations to do so.

For projects that are not required to assess permanence risk, complete the table below for the project crediting period.

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
DD-MMM-YYYY to 31-Dec-YYYY	Example: 50,000	Example: 20,000	Example: 10,000	Example: 10,000	Example: 10,000	Example: 20,000
01-Jan-YYYY to 31-Dec-YYYY						
01-Jan-YYYY to DD-MMM-YYYY						

Total						
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For projects required to assess permanence risk, complete the table below for the project crediting period.

Note that the buffer pool allocation is split proportionally between the estimated reductions and removals. (For example, if a project is estimated to achieve 20,000 tCO₂e removals and 80,000 tCO₂e reductions and has a buffer contribution of 20%, or 20,000, the estimated removal VCUs would be 16,000 and reduction VCUs would be 64,000).

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated buffer pool allocation (tCO ₂ e)	Estimated reduction VCUs (tCO ₂ e)	Estimated removal VCUs (tCO ₂ e)	Estimated total VCU issuance (tCO ₂ e)
DD-MMM-YYYY to 31-Dec-YYYY	Example: 50,000	Example: 20,000	Example: 10,000	Example: 4,000	Example: 8,000	Example: 8,000	Example: 16,000
01-Jan-YYYY to 31-Dec-YYYY							
01-Jan-YYYY to DD-MMM-YYYY							
Total							

6 MONITORING

6.1 Data and Parameters Available at Validation

Complete the table below for all data and parameters that are determined or available at validation and remain fixed throughout the project crediting period (copy the table as necessary for each data/parameter). The values provided are used to quantify the estimated reductions and removals for the project crediting period in Section 5 above. Data and parameters monitored during the operation of the project are included in Section 6.2 (Data and Parameters Monitored) below.

Data / Parameter	
------------------	--

Data unit	<i>Indicate the unit of measure</i>
Description	<i>Provide a brief description of the data/parameter</i>
Source of data	<i>Indicate the source(s) of data</i>
Value applied:	<i>Provide the value applied</i>
Justification of choice of data or description of measurement methods and procedures applied	<i>Justify the choice of data source, providing references where applicable. Where values are based on measurement, include a description of the measurement methods and procedures applied (e.g., what standards or protocols have been followed), indicate the responsible person/entity that undertook the measurement, the date of the measurement and the measurement results. More detailed information may be provided in an appendix.</i>
Purpose of data	<i>Indicate one of the following:</i> <i>Determination of baseline scenario (AFOLU projects only)</i> <i>Calculation of baseline emissions</i> <i>Calculation of project emissions</i> <i>Calculation of leakage</i>
Comments	<i>Provide any additional comments</i>

6.2 Data and Parameters Monitored

Complete the table below for all data and parameters to be monitored during the project crediting period (copy the table as necessary for each data/parameter). The values provided are used to estimate the reductions and removals for the project crediting period in Section 5 above. Data and parameters determined or available at validation are included in Section 6.1 (Data and Parameters Available at Validation) above.

Data / Parameter	
Data unit	<i>Indicate the unit of measure</i>
Description	<i>Provide a brief description of the data/parameter</i>
Source of data	<i>Indicate the source(s) of data</i>

Description of measurement methods and procedures applied	<i>Specify the measurement methods and procedures, any standards or protocols followed, and the person/entity responsible for the measurement. Include any relevant information regarding the accuracy of the measurements (e.g., accuracy associated with meter equipment or laboratory tests).</i>
Frequency of monitoring/recording	<i>Specify measurement and recording frequency</i>
Value applied:	<i>Provide an estimated value for the data/parameter</i>
Monitoring equipment	<i>Identify equipment used to monitor the data/parameter including type, accuracy class, and serial number of equipment, as appropriate.</i>
QA/QC procedures applied	<i>Describe the quality assurance and quality control (QA/QC) procedures applied, including the calibration procedures where applicable.</i>
Purpose of data	<i>Indicate one of the following:</i> <i>Calculation of baseline emissions</i> <i>Calculation of project emissions</i> <i>Calculation of leakage</i>
Calculation method	<i>Where relevant, provide the calculation method, including any equations, used to establish the data/parameter.</i>
Comments	<i>Provide any additional comments</i>

6.3 Monitoring Plan

Describe the process and schedule for obtaining, compiling, and analyzing the monitored data and parameters set out in Section 6.2 (Data and Parameters Monitored) above.

Include details on the following:

- The methods used for measuring, recording, storing, aggregating, collating and reporting on monitored data and parameters. Where relevant, include the processes used for calibrating monitoring equipment.*
- The organizational structure, responsibilities and competencies of the personnel that carried out monitoring activities.*
- The procedures for internal auditing and QA/QC.*
- The procedures for handling non-conformances with the validated monitoring plan.*

- Any sampling approaches used, including target precision levels, sample sizes, sample site locations, stratification, frequency of measurement and QA/QC procedures.

Where appropriate, include line diagrams to display the GHG data collection and management system.

7 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

7.1 Data and Parameters Monitored

Complete the table below for all data and parameters monitored during the monitoring period (copy the table as necessary for each data/parameter). The values provided are used to quantify actual reductions and removals achieved for the monitoring period. Data and parameters determined or available at validation which remain fixed throughout the project crediting period are included in Section 6.1 (Data and Parameters Available at Validation) above.

Data / Parameter	
Data unit	Indicate the unit of measure
Description	Provide a brief description of the data/parameter
Value applied:	Provide the monitored value for the data/parameter
Comments	Provide any additional comments

7.2 Baseline Emissions

Quantify the baseline emissions and/or carbon stock changes for the monitoring period in accordance with the applied methodology. Baseline emissions may be negative where carbon stock increases (sinks) exceed baseline emissions. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculation. Include all calculations in the emission reduction and removal calculation spreadsheet.

7.3 Project Emissions

Quantify project emission and/or carbon stock changes for the monitoring period in accordance with the applied methodology. Project emissions may be negative where carbon

stock increases (sinks) exceed project emissions Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculation. Include all calculations in the emission reduction and removal calculation spreadsheet.

7.4 Leakage Emissions

Quantify leakage emissions for the monitoring period in accordance with the applied methodology. Specify the reductions and removals separately where the applied methodology provides procedures and equations to do so. Include all relevant equations here and provide sufficient information to allow the reader to reproduce the calculation. Include all calculations in the emission reduction and removal calculation spreadsheet.

7.5 GHG Emission Reductions and Carbon Dioxide Removals

Quantify the GHG emission reductions (reductions) and carbon dioxide removals (removals) for the monitoring period. Include all relevant equations.

Complete the tables below by vintage period (calendar year). Note that the baseline or project emissions subtotals may be negative where sinks exceed emissions. Only specify the estimated VCUUs for reductions and removals separately where the applied methodology provides procedures and equations to do so.

For projects that are not required to assess permanence risk, complete the table below for the project crediting period:

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCUUs (tCO ₂ e)	Removal VCUUs (tCO ₂ e)	Total VCUUs (tCO ₂ e)
DD-MMM-YYYY to 31-Dec-YYYY	Example: 50,000	Example: 20,000	Example: 10,000	Example: 10,000	Example: 10,000	Example: 20,000
01-Jan-YYYY to 31-Dec-YYYY						
01-Jan-YYYY to DD-MMM-YYYY						
...						
Total						

For projects required to assess permanence risk:

i) Provide the requested information using the table below:

State the non-permanence risk rating (%)	
Has the non-permanence risk report been attached as either an appendix or a separate document?	☐ Yes ☐ No
For ARR and IFM projects with harvesting, state, in tCO ₂ e the Long-term Average (LTA).	
Has the LTA been updated based on monitored data, if applicable?	☐ Yes ☐ No If no, provide justification.
State, in tCO ₂ e, the expected total GHG benefit to date	
If a loss occurred (including a loss event or reversal), state the amount of tCO ₂ e lost:	

ii) Complete the table below for the project crediting period. Note that the buffer pool allocation is split proportionally between the reductions and removals. (For example, if a project achieves 20,000 tCO₂e removals and 80,000 tCO₂e reductions and has a buffer contribution of 20%, or 20,000, the removal VCUs would be 16,000 and reduction VCUs would be 64,000).

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Buffer pool allocation (tCO ₂ e)	Reductions VCUs (tCO ₂ e)	Removals VCUs (tCO ₂ e)	Total VCU issuance (tCO ₂ e)
DD-MMM-YYYY to 31-Dec-YYYY	Example: 50,000	Example: 20,000	Example: 10,000	Example: 4,000	Example: 8,000	Example: 8,000	Example: 16,000
01-Jan-YYYY to 31-Dec-YYYY							
01-Jan-YYYY to DD-MMM-YYYY							
Total							

For all projects, state the estimated ex-ante GHG emission reductions and carbon dioxide removals and the achieved reductions and removals for the monitoring period. Report the percentage difference and explain any difference. The quantities of reductions and removals are the total quantities before any deductions for buffer credits.

Vintage period	Ex-ante estimated reductions/ removals	Achieved reductions/ removals	Percent difference	Explanation for the difference
DD-MMM-YYYY to 31-Dec-YYYY				
01-Jan-YYYY to 31-Dec-YYYY				
...				
01-Jan-YYYY to DD-MMM-YYYY				
Total				

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APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Use the table below to describe the commercially sensitive information included in the project description to be excluded in the public version.

Section	Information	Justification

APPENDIX

Annex 001.	DECREE No. 246, OF OCTOBER 29, 1991 CANA BRAVA
Annex 002.	DECREE No. 382 OF DECEMBER 24, 1991 URUCU JURUA
Annex 003.	DECREE OF FEBRUARY 10, 2003 URUCU JURUA
Annex 004.	DECREE No. 313 OF October 29, 1991 Lagoa Comprida
Annex 005.	DECREE No. 88,813, OF OCTOBER 4, 1983 RODEADOR
Annex 006a	VILLAGES TI CANABRAVA REGIÃO DO MEARIM
Annex 006b.	VILLAGES TI CANABRAVA - REGIÃO ILHA SÃO PEDRO E CACIMBA VELHO
Annex 006c.	VILLAGES TI CANABRAVA RIO CORDA
Annex 006d.	VILLAGES TI CANABRAVA REGIÃO DO - MA 328
Annex 006e.	VILLAGES TI CANABRAVA BRAÇO DO CACHOEIRA MEARIM
Annex 006f.	VILLAGES BR-226 TI CANABRAVA - BR
Annex 007.	VILLAGES TI LAGOA COMPRIDA
Annex 008.	VILLAGES TI RODEADOR
Annex 009.	VILLAGES TI URUCU JURUA
Annex 010.	INTENTION LETTER
Annex 011.	SUPREME COUNCIL STATUS
Annex 012.	Partnership and Cooperation Agreement for Ka'a Pyahu project - Legal Design
Annex 013.	MEETING MINUTES PREVIOUS CONSULTATION
Annex 014.	OF,N01-2023 INDIGENOUS HEALTH
Annex 015.	HUMAN RIGHTS SECRETARIAT
Annex 016.	MEDICAL AND HEALTH TREATMENTS
Annex 017.	PROC ADM FUNAI
Annex 017a.	CE-PR-0077-2023 Reiteration of emergency operational maintenance request and continuation of negotiations accordingly
Annex 017b.	CE PR 0072-2023 -Guajajara Transition Program for PBA-CI - LT kV Tucuruí - Marabá - Imperatriz - P.Dutra - S.Luís II (C1 and C2) 4
Annex 018.	Term of Commitment_FUNAI TRANSMISSION LINE
Annex 018a.	MINUTES GENERAL ASSEMBLY JUDICIAL PROCESS COMPENSATORY MEASURES
Annex 018b.	MINUTES ALDEIA COQUINHO MEETING
Annex 018c.	C.S Caciques para Antônio Augusto B.P
Annex 018d.	INSTRUMENTAL APPEAL
Annex 018e.	JUDICIAL MANIFESTATION IN DEFENSE OF THE TERRITORIES
Annex 018f.	Draft Conduct Adjustment Term - Supreme Council
Annex 018g.	OF - DR HILTON - MPF
Annex 018h.	OF COUNCIL REQUESTING ACTIONS FROM ELETRONORTE
Annex 018i.	Of.Pres. 038-2023- AP -Transmission lines - JOSÉ LUCIANO CLEMENTE GUAJAJARA
Annex 018j.	REQUEST FOR INTERVENTION FROM THE REGIONAL ATTORNEY OF THE REPUBLIC
Annex 019.	Report-violence-povos-indigenas-2022-cimi

